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Inventor(s) DUAN; Weimin et al.

SOUNDING REFERENCE SIGNAL DESIGN FOR MULTIPLE USE CASES

Abstract

Disclosed are techniques for wireless communication. In an aspect, a user equipment (UE) transmits one or more sounding reference signal (SRS) resources, wherein a waveform of the one or more SRS resources is an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform.

Inventors: **DUAN; Weimin (San Diego, CA), LIU; Kangqi (San Diego, CA)**

Applicant: **QUALCOMM Incorporated** (San Diego, CA)

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Background/Summary

BACKGROUND OF THE DISCLOSURE
1. Field of the Disclosure
[0001] Aspects of the disclosure relate generally to wireless technologies.
2. Description of the Related Art

[0002] Wireless communication systems have developed through various generations, including a first-generation analog wireless phone service (1G), a second-generation (2G) digital wireless phone service (including interim 2.5G and 2.75G networks), a third-generation (3G) high speed data, Internet-capable wireless service and a fourth-generation (4G) service (e.g., Long Term Evolution (LTE) or WiMax). There are presently many different types of wireless communication systems in use, including cellular and personal communications service (PCS) systems. Examples of known cellular systems include the cellular analog advanced mobile phone system (AMPS), and digital cellular systems based on code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), the Global System for Mobile communications (GSM), etc.
[0003] A fifth generation (5G) wireless standard, referred to as New Radio (NR), enables higher data transfer speeds, greater numbers of connections, and better coverage, among other improvements. The 5G standard, according to the Next Generation Mobile Networks Alliance, is designed to provide higher data rates as compared to previous standards, more accurate positioning (e.g., based on reference signals for positioning (RS-P), such as downlink, uplink, or sidelink positioning reference signals (PRS)), and other technical enhancements. These enhancements, as well as the use of higher frequency bands, advances in PRS processes and technology, and high-density deployments for 5G, enable highly accurate 5G-based positioning.
SUMMARY
[0004] The following presents a simplified summary relating to one or more aspects disclosed herein. Thus, the following summary should not be considered an extensive overview relating to all contemplated aspects, nor should the following summary be considered to identify key or critical elements relating to all contemplated aspects or to delineate the scope associated with any particular aspect. Accordingly, the following summary has the sole purpose to present certain concepts relating to one or more aspects relating to the mechanisms disclosed herein in a simplified form to precede the detailed description presented below.
[0005] In an aspect, a method of wireless communication performed by a user equipment (UE) includes transmitting one or more sounding reference signal (SRS) resources, wherein a waveform of the one or more SRS resources is an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform.
[0006] In an aspect, a method of communication performed by a network entity transmits, to a user equipment (UE), a request for capabilities of the UE related to transmission of sounding reference signal (SRS) resources by the UE; and receives, from the UE, one or more capabilities of the UE in response to the request, wherein the one or more capabilities of the UE indicate whether the UE supports an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform for the transmission of SRS resources.
[0007] In an aspect, a user equipment (UE) includes one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: transmit, via the one or more transceivers, one or more sounding reference signal (SRS) resources, wherein a waveform of the one or more SRS resources is an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform.
[0008] In an aspect, a network entity includes one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: transmit, via the one or more transceivers, to a user equipment (UE), a request for capabilities of the UE related to transmission of sounding reference signal (SRS) resources by the UE; and receive, via the one or more transceivers, from the UE, one or more capabilities of the UE in response to the request, wherein the one or more capabilities of the UE indicate whether the UE supports an

orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform for the transmission of SRS resources.
[0009] In an aspect, a user equipment (UE) includes means for transmitting one or more sounding reference signal (SRS) resources, wherein a waveform of the one or more SRS resources is an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform.
[0010] In an aspect, a network entity includes means for transmitting, to a user equipment (UE), a request for capabilities of the UE related to transmission of sounding reference signal (SRS) resources by the UE; and means for receiving, from the UE, one or more capabilities of the UE in response to the request, wherein the one or more capabilities of the UE indicate whether the UE supports an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform for the transmission of SRS resources.
[0011] In an aspect, a non-transitory computer-readable medium stores computer-executable instructions that, when executed by a user equipment (UE), cause the UE to: transmit one or more sounding reference signal (SRS) resources, wherein a waveform of the one or more SRS resources is an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform.
[0012] In an aspect, a non-transitory computer-readable medium stores computer-executable instructions that, when executed by a network entity, cause the network entity to: transmit, to a user equipment (UE), a request for capabilities of the UE related to transmission of sounding reference signal (SRS) resources by the UE; and receive, from the UE, one or more capabilities of the UE in response to the request, wherein the one or more capabilities of the UE indicate whether the UE supports an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform for the transmission of SRS resources.
[0013] Other objects and advantages associated with the aspects disclosed herein will be apparent to those skilled in the art based on the accompanying drawings and detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0014] The accompanying drawings are presented to aid in the description of various aspects of the disclosure and are provided solely for illustration of the aspects and not limitation thereof.
[0015] FIG. **1** illustrates an example wireless communications system, according to aspects of the disclosure.
[0016] FIGS. **2**A and **2**B illustrate example wireless network structures, according to aspects of the disclosure.
[0017] FIGS. **3**A, **3**B, and **3**C are simplified block diagrams of several sample aspects of components that may be employed in a user equipment (UE), a base station, and a network entity, respectively, and configured to support communications as taught herein.
[0018] FIG. **4** illustrates examples of various positioning methods supported in New Radio (NR), according to aspects of the disclosure.
[0019] FIGS. **5**A and **5**B illustrate different types of wireless sensing, according to aspects of the disclosure.
[0020] FIG. **6** is a diagram illustrating an example frame structure, according to aspects of the disclosure.
[0021] FIGS. **7**A to **7**C illustrate various comb patterns supported for SRS-for-positioning within a resource block, according to aspects of the disclosure.
[0022] FIGS. **8** and **9** illustrate graphs of example frequency modulated continuous wave (FMCW) waveforms, according to aspects of the disclosure.

[0023] FIG. **10** illustrates an example FMCW system, according to aspects of the disclosure.
[0024] FIG. **11** illustrates the two types of cyclic prefix orthogonal frequency division multiplexing (CP-OFDM) compatible FMCW waveforms for analog transmission and/or sensing, according to aspects of the disclosure.
[0025] FIG. **12** illustrates a comparison of analog versus digital generation of a Type 2 FMCW, according to aspects of the disclosure.
[0026] FIG. **13** illustrates the example frequency division multiplexing (FDM) designs, according to aspects of the disclosure.
[0027] FIG. **14** illustrates a comparison of back-to-back staggered sounding reference signals (SRS) without gaps and staggered SRS with gaps, according to aspects of the disclosure.
[0028] FIG. **15** is a diagram illustrating an example of different bandwidth part (BWP) switching latencies, according to aspects of the disclosure.
[0029] FIGS. **16**A and **16**B illustrate various concepts of Doppler division multiplexing (DDM), according to aspects of the disclosure.
[0030] FIGS. **17**A and **17**B illustrate an example of resource level DDM, according to aspects of the disclosure.
[0031] FIG. **18** illustrates an example method of wireless communication, according to aspects of the disclosure.
[0032] FIG. **19** illustrates an example method of communication, according to aspects of the disclosure.

DETAILED DESCRIPTION

[0033] Aspects of the disclosure are provided in the following description and related drawings directed to various examples provided for illustration purposes. Alternate aspects may be devised without departing from the scope of the disclosure. Additionally, well-known elements of the disclosure will not be described in detail or will be omitted so as not to obscure the relevant details of the disclosure.
[0034] Various aspects relate generally to wireless communication. Some aspects more specifically relate to sounding reference signal (SRS) design for multiple use cases, such as channel sounding, wireless positioning, and environment sensing. In some examples, the proposed SRS designs consider FMCW compatibility, BWP issues, and Doppler division multiplexing.
[0035] Particular aspects of the subject matter described in this disclosure can be implemented to realize one or more of the following potential advantages. In some examples, by providing an SRS design for multiple use cases, the described techniques can be used to improve uplink communication coverage, positioning and sensing capacity, and power consumption.
[0036] The words “exemplary” and/or “example” are used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” and/or “example” is not necessarily to be construed as preferred or advantageous over other aspects. Likewise, the term “aspects of the disclosure” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation.
[0037] Those of skill in the art will appreciate that the information and signals described below may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description below may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof, depending in part on the particular application, in part on the desired design, in part on the corresponding technology, etc.
[0038] Further, many aspects are described in terms of sequences of actions to be performed by, for example, elements of a computing device. It will be recognized that various actions described herein can be performed by specific circuits (e.g., application specific integrated circuits (ASICs)), by program instructions being executed by one or more processors, or by a combination of both.

Additionally, the sequence(s) of actions described herein can be considered to be embodied entirely within any form of non-transitory computer-readable storage medium having stored therein a corresponding set of computer instructions that, upon execution, would cause or instruct an associated processor of a device to perform the functionality described herein. Thus, the various aspects of the disclosure may be embodied in a number of different forms, all of which have been contemplated to be within the scope of the claimed subject matter. In addition, for each of the aspects described herein, the corresponding form of any such aspects may be described herein as, for example, “logic configured to” perform the described action.

[0039] As used herein, the terms “user equipment” (UE) and “base station” are not intended to be specific or otherwise limited to any particular radio access technology (RAT), unless otherwise noted. In general, a UE may be any wireless communication device (e.g., a mobile phone, router, tablet computer, laptop computer, consumer asset locating device, wearable (e.g., smartwatch, glasses, augmented reality (AR)/virtual reality (VR) headset, etc.), vehicle (e.g., automobile, motorcycle, bicycle, etc.), Internet of Things (IoT) device, etc.) used by a user to communicate over a wireless communications network. A UE may be mobile or may (e.g., at certain times) be stationary, and may communicate with a radio access network (RAN). As used herein, the term “UE” may be referred to interchangeably as an “access terminal” or “AT,” a “client device,” a “wireless device,” a “subscriber device,” a “subscriber terminal,” a “subscriber station,” a “user terminal” or “UT,” a “mobile device,” a “mobile terminal,” a “mobile station,” or variations thereof. Generally, UEs can communicate with a core network via a RAN, and through the core network the UEs can be connected with external networks such as the Internet and with other UEs. Of course, other mechanisms of connecting to the core network and/or the Internet are also possible for the UEs, such as over wired access networks, wireless local area network (WLAN) networks (e.g., based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 specification, etc.) and so on.

[0040] A base station may operate according to one of several RATs in communication with UEs depending on the network in which it is deployed, and may be alternatively referred to as an access point (AP), a network node, a NodeB, an evolved NodeB (eNB), a next generation cNB (ng-eNB), a New Radio (NR) Node B (also referred to as a gNB or gNodeB), etc. A base station may be used primarily to support wireless access by UEs, including supporting data, voice, and/or signaling connections for the supported UEs. In some systems a base station may provide purely edge node signaling functions while in other systems it may provide additional control and/or network management functions. A communication link through which UEs can send signals to a base station is called an uplink (UL) channel (e.g., a reverse traffic channel, a reverse control channel, an access channel, etc.). A communication link through which the base station can send signals to UEs is called a downlink (DL) or forward link channel (e.g., a paging channel, a control channel, a broadcast channel, a forward traffic channel, etc.). As used herein the term traffic channel (TCH) can refer to either an uplink/reverse or downlink/forward traffic channel.

[0041] The term “base station” may refer to a single physical transmission-reception point (TRP) or to multiple physical TRPs that may or may not be co-located. For example, where the term “base station” refers to a single physical TRP, the physical TRP may be an antenna of the base station corresponding to a cell (or several cell sectors) of the base station. Where the term “base station” refers to multiple co-located physical TRPs, the physical TRPs may be an array of antennas (e.g., as in a multiple-input multiple-output (MIMO) system or where the base station employs beamforming) of the base station. Where the term “base station” refers to multiple non-co-located physical TRPs, the physical TRPs may be a distributed antenna system (DAS) (a network of spatially separated antennas connected to a common source via a transport medium) or a remote radio head (RRH) (a remote base station connected to a serving base station). Alternatively, the non-co-located physical TRPs may be the serving base station receiving the measurement report from the UE and a neighbor base station whose reference radio frequency (RF) signals the UE is measuring. Because a TRP is the point from which a base station transmits and receives wireless signals, as used herein, references to transmission from or reception at a base station are to be understood as referring to a particular TRP of the base station.

[0042] In some implementations that support positioning of UEs, a base station may not support wireless access by UEs (e.g., may not support data, voice, and/or signaling connections for UEs), but may instead transmit reference signals to UEs to be measured by the UEs, and/or may receive and measure signals transmitted by the UEs. Such a base station may be referred to as a positioning beacon (e.g., when transmitting signals to UEs) and/or as a location measurement unit (e.g., when receiving and measuring signals from UEs).

[0043] An “RF signal” comprises an electromagnetic wave of a given frequency that transports information through the space between a transmitter and a receiver. As used herein, a transmitter may transmit a single “RF signal” or multiple “RF signals” to a receiver. However, the receiver may receive multiple “RF signals” corresponding to each transmitted RF signal due to the propagation characteristics of RF signals through multipath channels. The same transmitted RF signal on different paths between the transmitter and receiver may be referred to as a “multipath” RF signal. As used herein, an RF signal may also be referred to as a “wireless signal” or simply a “signal” where it is clear from the context that the term “signal” refers to a wireless signal or an RF signal.

[0044] FIG. **1** illustrates an example wireless communications system **100**, according to aspects of the disclosure. The wireless communications system **100** (which may also be referred to as a wireless wide area network (WWAN)) may include various base stations **102** (labeled “BS”) and various UEs **104**. The base stations **102** may include macro cell base stations (high power cellular base stations) and/or small cell base stations (low power cellular base stations). In an aspect, the macro cell base stations may include eNBs and/or ng-eNBs where the wireless communications system **100** corresponds to an LTE network, or gNBs where the wireless communications system **100** corresponds to a NR network, or a combination of both, and the small cell base stations may include femtocells, picocells, microcells, etc.

[0045] The base stations **102** may collectively form a RAN and interface with a core network **170** (e.g., an evolved packet core (EPC) or a 5G core (5GC)) through backhaul links **122**, and through the core network **170** to one or more location servers **172** (e.g., a location management function (LMF) or a secure user plane location (SUPL) location platform (SLP)). The location server(s) **172** may be part of core network **170** or may be external to core network **170**. A location server **172** may be integrated with a base station **102**. A UE **104** may communicate with a location server **172** directly or indirectly. For example, a UE **104** may communicate with a location server **172** via the base station **102** that is currently serving that UE **104**. A UE **104** may also communicate with a location server **172** through another path, such as via an application server (not shown), via another network, such as via a wireless local area network (WLAN) access point (AP) (e.g., AP **150** described below), and so on. For signaling purposes, communication between a UE **104** and a location server **172** may be represented as an indirect connection (e.g., through the core network **170**, etc.) or a direct connection (e.g., as shown via direct connection **128**), with the intervening nodes (if any) omitted from a signaling diagram for clarity.

[0046] In addition to other functions, the base stations **102** may perform functions that relate to one or more of transferring user data, radio channel ciphering and deciphering, integrity protection, header compression, mobility control functions (e.g., handover, dual connectivity), inter-cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, RAN sharing, multimedia broadcast multicast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, and delivery of warning messages. The base stations **102** may communicate with each other directly or indirectly (e.g., through the EPC/5GC) over backhaul links **134**, which may be wired or wireless.

[0047] The base stations **102** may wirelessly communicate with the UEs **104**. Each of the base stations **102** may provide communication coverage for a respective geographic coverage area **110**. In an aspect, one or more cells may be supported by a base station **102** in each geographic coverage area **110**. A "cell" is a logical communication entity used for communication with a base station (e.g., over some frequency resource, referred to as a carrier frequency, component carrier, carrier, band, or the like), and may be associated with an identifier (e.g., a physical cell identifier (PCI), an enhanced cell identifier (ECI), a virtual cell identifier (VCI), a cell global identifier (CGI), etc.) for distinguishing cells operating via the same or a different carrier frequency. In some cases, different cells may be configured according to different protocol types (e.g., machine-type communication (MTC), narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB), or others) that may provide access for different types of UEs. Because a cell is supported by a specific base station, the term "cell" may refer to either or both of the logical communication entity and the base station that supports it, depending on the context. In addition, because a TRP is typically the physical transmission point of a cell, the terms "cell" and "TRP" may be used interchangeably. In some cases, the term "cell" may also refer to a geographic coverage area of a base station (e.g., a sector), insofar as a carrier frequency can be detected and used for communication within some portion of geographic coverage areas **110**.

[0048] While neighboring macro cell base station **102** geographic coverage areas **110** may partially overlap (e.g., in a handover region), some of the geographic coverage areas **110** may be substantially overlapped by a larger geographic coverage area **110**. For example, a small cell base station **102′** (labeled "SC" for "small cell") may have a geographic coverage area **110′** that substantially overlaps with the geographic coverage area **110** of one or more macro cell base stations **102**. A network that includes both small cell and macro cell base stations may be known as a heterogeneous network. A heterogeneous network may also include home eNBs (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG).

[0049] The communication links **120** between the base stations **102** and the UEs **104** may include uplink (also referred to as reverse link) transmissions from a UE **104** to a base station **102** and/or downlink (DL) (also referred to as forward link) transmissions from a base station **102** to a UE **104**. The communication links **120** may use MIMO antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links **120** may be through one or more carrier frequencies. Allocation of carriers may be asymmetric with respect to downlink and uplink (e.g., more or less carriers may be allocated for downlink than for uplink).

[0050] The wireless communications system **100** may further include a wireless local area network (WLAN) access point (AP) **150** in communication with WLAN stations (STAs) **152** via communication links **154** in an unlicensed frequency spectrum (e.g., 5 GHZ). When communicating in an unlicensed frequency spectrum, the WLAN STAs **152** and/or the WLAN AP **150** may perform a clear channel assessment (CCA) or listen before talk (LBT) procedure prior to communicating in order to determine whether the channel is available.

[0051] The small cell base station **102′** may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell base station **102′** may employ LTE or NR technology and use the same 5 GHz unlicensed frequency spectrum as used by the WLAN AP **150**. The small cell base station **102′**, employing LTE/5G in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network. NR in unlicensed spectrum may be referred to as NR-U. LTE in an unlicensed spectrum may be referred to as LTE-U, licensed assisted access (LAA), or MULTEFIRE®.

[0052] The wireless communications system **100** may further include a millimeter wave (mmW) base station **180** that may operate in mmW frequencies and/or near mmW frequencies in communication with a UE **182**. Extremely high frequency (EHF) is part of the RF in the electromagnetic spectrum. EHF has a range of 30 GHz to 300 GHz and a wavelength between 1 millimeter and 10 millimeters. Radio waves in this band may be referred to as a millimeter wave.

Near mmW may extend down to a frequency of 3 GHZ with a wavelength of 100 millimeters. The super high frequency (SHF) band extends between 3 GHz and 30 GHz, also referred to as centimeter wave. Communications using the mmW/near mmW radio frequency band have high path loss and a relatively short range. The mmW base station **180** and the UE **182** may utilize beamforming (transmit and/or receive) over a mmW communication link **184** to compensate for the extremely high path loss and short range. Further, it will be appreciated that in alternative configurations, one or more base stations **102** may also transmit using mmW or near mmW and beamforming. Accordingly, it will be appreciated that the foregoing illustrations are merely examples and should not be construed to limit the various aspects disclosed herein.

[0053] Transmit beamforming is a technique for focusing an RF signal in a specific direction. Traditionally, when a network node (e.g., a base station) broadcasts an RF signal, it broadcasts the signal in all directions (omni-directionally). With transmit beamforming, the network node determines where a given target device (e.g., a UE) is located (relative to the transmitting network node) and projects a stronger downlink RF signal in that specific direction, thereby providing a faster (in terms of data rate) and stronger RF signal for the receiving device(s). To change the directionality of the RF signal when transmitting, a network node can control the phase and relative amplitude of the RF signal at each of the one or more transmitters that are broadcasting the RF signal. For example, a network node may use an array of antennas (referred to as a “phased array” or an “antenna array”) that creates a beam of RF waves that can be “steered” to point in different directions, without actually moving the antennas. Specifically, the RF current from the transmitter is fed to the individual antennas with the correct phase relationship so that the radio waves from the separate antennas add together to increase the radiation in a desired direction, while cancelling to suppress radiation in undesired directions.

[0054] Transmit beams may be quasi-co-located, meaning that they appear to the receiver (e.g., a UE) as having the same parameters, regardless of whether or not the transmitting antennas of the network node themselves are physically co-located. In NR, there are four types of quasi-co-location (QCL) relations. Specifically, a QCL relation of a given type means that certain parameters about a second reference RF signal on a second beam can be derived from information about a source reference RF signal on a source beam. Thus, if the source reference RF signal is QCL Type A, the receiver can use the source reference RF signal to estimate the Doppler shift, Doppler spread, average delay, and delay spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type B, the receiver can use the source reference RF signal to estimate the Doppler shift and Doppler spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type C, the receiver can use the source reference RF signal to estimate the Doppler shift and average delay of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type D, the receiver can use the source reference RF signal to estimate the spatial receive parameter of a second reference RF signal transmitted on the same channel.

[0055] In receive beamforming, the receiver uses a receive beam to amplify RF signals detected on a given channel. For example, the receiver can increase the gain setting and/or adjust the phase setting of an array of antennas in a particular direction to amplify (e.g., to increase the gain level of) the RF signals received from that direction. Thus, when a receiver is said to beamform in a certain direction, it means the beam gain in that direction is high relative to the beam gain along other directions, or the beam gain in that direction is the highest compared to the beam gain in that direction of all other receive beams available to the receiver. This results in a stronger received signal strength (e.g., reference signal received power (RSRP), reference signal received quality (RSRQ), signal-to-interference-plus-noise ratio (SINR), etc.) of the RF signals received from that direction.

[0056] Transmit and receive beams may be spatially related. A spatial relation means that parameters for a second beam (e.g., a transmit or receive beam) for a second reference signal can be derived from information about a first beam (e.g., a receive beam or a transmit beam) for a first reference signal. For example, a UE may use a particular receive beam to receive a reference downlink reference signal (e.g., synchronization signal block (SSB)) from a base station. The UE can then form a transmit beam for sending an uplink reference signal (e.g., sounding reference signal (SRS)) to that base station based on the parameters of the receive beam.

[0057] Note that a "downlink" beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is forming the downlink beam to transmit a reference signal to a UE, the downlink beam is a transmit beam. If the UE is forming the downlink beam, however, it is a receive beam to receive the downlink reference signal. Similarly, an "uplink" beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is forming the uplink beam, it is an uplink receive beam, and if a UE is forming the uplink beam, it is an uplink transmit beam.

[0058] The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHZ) and FR2 (24.25 GHz-52.6 GHz). It should be understood that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a "Sub-6 GHz" band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a "millimeter wave" band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHZ-300 GHz) which is identified by the INTERNATIONAL TELECOMMUNICATION UNION® as a "millimeter wave" band.

[0059] The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHZ-24.25 GHZ). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHZ-300 GHz). Each of these higher frequency bands falls within the EHF band.

[0060] With the above aspects in mind, unless specifically stated otherwise, it should be understood that the term "sub-6 GHz" or the like if used herein may broadly represent frequencies that may be less than 6 GHZ, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term "millimeter wave" or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band.

[0061] In a multi-carrier system, such as 5G, one of the carrier frequencies is referred to as the "primary carrier" or "anchor carrier" or "primary serving cell" or "PCell," and the remaining carrier frequencies are referred to as "secondary carriers" or "secondary serving cells" or "SCells." In carrier aggregation, the anchor carrier is the carrier operating on the primary frequency (e.g., FR1) utilized by a UE **104**/**182** and the cell in which the UE **104**/**182** either performs the initial radio resource control (RRC) connection establishment procedure or initiates the RRC connection re-establishment procedure. The primary carrier carries all common and UE-specific control channels, and may be a carrier in a licensed frequency (however, this is not always the case). A secondary carrier is a carrier operating on a second frequency (e.g., FR2) that may be configured once the RRC connection is established between the UE **104** and the anchor carrier and that may be used to provide additional radio resources. In some cases, the secondary carrier may be a carrier in an unlicensed frequency. The secondary carrier may contain only necessary signaling information and signals, for example, those that are UE-specific may not be present in the secondary carrier, since both primary uplink and downlink carriers are typically UE-specific. This means that different UEs **104**/**182** in a cell may have different downlink primary carriers. The same is true for the uplink primary carriers. The network is able to change the primary carrier of any UE **104**/**182** at any time. This is done, for example, to balance the load on different carriers. Because a "serving cell" (whether a PCell or an SCell) corresponds to a carrier frequency/component carrier over which some base station is communicating, the term "cell," "serving cell," "component carrier," "carrier frequency," and the like can be used interchangeably.

[0062] For example, still referring to FIG. **1**, one of the frequencies utilized by the macro cell base stations **102** may be an anchor carrier (or "PCell") and other frequencies utilized by the macro cell base stations **102** and/or the mmW base station **180** may be secondary carriers ("SCells"). The simultaneous transmission and/or reception of multiple carriers enables the UE **104**/**182** to significantly increase its data transmission and/or reception rates. For example, two 20 MHz aggregated carriers in a multi-carrier system would theoretically lead to a two-fold increase in data rate (i.e., 40 MHZ), compared to that attained by a single 20 MHz carrier.

[0063] The wireless communications system **100** may further include a UE **164** that may communicate with a macro cell base station **102** over a communication link **120** and/or the mmW base station **180** over a mmW communication link **184**. For example, the macro cell base station **102** may support a PCell and one or more SCells for the UE **164** and the mmW base station **180** may support one or more SCells for the UE **164**.

[0064] In some cases, the UE **164** and the UE **182** may be capable of sidelink communication. Sidelink-capable UEs (SL-UEs) may communicate with base stations **102** over communication links **120** using the Uu interface (i.e., the air interface between a UE and a base station). SL-UEs (e.g., UE **164**, UE **182**) may also communicate directly with each other over a wireless sidelink **160** using the PC5 interface (i.e., the air interface between sidelink-capable UEs). A wireless sidelink (or just "sidelink") is an adaptation of the core cellular (e.g., LTE, NR) standard that allows direct communication between two or more UEs without the communication needing to go through a base station. Sidelink communication may be unicast or multicast, and may be used for device-to-device (D2D) media-sharing, vehicle-to-vehicle (V2V) communication, vehicle-to-everything (V2X) communication (e.g., cellular V2X (cV2X) communication, enhanced V2X (eV2X) communication, etc.), emergency rescue applications, etc. One or more of a group of SL-UEs utilizing sidelink communications may be within the geographic coverage area **110** of a base station **102**. Other SL-UEs in such a group may be outside the geographic coverage area **110** of a base station **102** or be otherwise unable to receive transmissions from a base station **102**. In some cases, groups of SL-UEs communicating via sidelink communications may utilize a one-to-many (1:M) system in which each SL-UE transmits to every other SL-UE in the group. In some cases, a base station **102** facilitates the scheduling of resources for sidelink communications. In other cases, sidelink communications are carried out between SL-UEs without the involvement of a base station **102**.

[0065] In an aspect, the sidelink **160** may operate over a wireless communication medium of interest, which may be shared with other wireless communications between other vehicles and/or infrastructure access points, as well as other RATs. A "medium" may be composed of one or more time, frequency, and/or space communication resources (e.g., encompassing one or more channels across one or more carriers) associated with wireless communication between one or more transmitter/receiver pairs. In an aspect, the medium of interest may correspond to at least a portion of an unlicensed frequency band shared among various RATs. Although different licensed frequency bands have been reserved for certain communication systems (e.g., by a government entity such as the Federal Communications Commission (FCC) in the United States), these systems, in particular those employing small cell access points, have recently extended operation into unlicensed frequency bands such as the Unlicensed National Information Infrastructure (U-NII) band used by wireless local area network (WLAN) technologies, most notably IEEE 802.11x WLAN technologies generally referred to as "Wi-Fi." Example systems of this type include different variants of CDMA systems, TDMA systems, FDMA systems, orthogonal FDMA (OFDMA) systems, single-carrier FDMA (SC-FDMA) systems, and so on.

[0066] Note that although FIG. **1** only illustrates two of the UEs as SL-UEs (i.e., UEs **164** and **182**), any of the illustrated UEs may be SL-UEs. Further, although only UE **182** was described as being capable of beamforming, any of the illustrated UEs, including UE **164**, may be capable of beamforming. Where SL-UEs are capable of beamforming, they may beamform towards each other (i.e., towards other SL-UEs), towards other UEs (e.g., UEs **104**), towards base stations (e.g., base stations **102**, **180**, small cell **102′**, access point **150**), etc. Thus, in some cases, UEs **164** and **182** may utilize beamforming over sidelink **160**.

[0067] In the example of FIG. **1**, any of the illustrated UEs (shown in FIG. **1** as a single UE **104** for simplicity) may receive signals **124** from one or more Earth orbiting space vehicles (SVs) **112** (e.g., satellites). In an aspect, the SVs **112** may be part of a satellite positioning system that a UE **104** can use as an independent source of location information. A satellite positioning system typically includes a system of transmitters (e.g., SVs **112**) positioned to enable receivers (e.g., UEs **104**) to determine their location on or above the Earth based, at least in part, on positioning signals (e.g., signals **124**) received from the transmitters. Such a transmitter typically transmits a signal marked with a repeating pseudo-random noise (PN) code of a set number of chips. While typically located in SVs **112**, transmitters may sometimes be located on ground-based control stations, base stations **102**, and/or other UEs **104**. A UE **104** may include one or more dedicated receivers specifically designed to receive signals **124** for deriving geo location information from the SVs **112**.

[0068] In a satellite positioning system, the use of signals **124** can be augmented by various satellite-based augmentation systems (SBAS) that may be associated with or otherwise enabled for use with one or more global and/or regional navigation satellite systems. For example an SBAS may include an augmentation system(s) that provides integrity information, differential corrections, etc., such as the Wide Area Augmentation System (WAAS), the European Geostationary Navigation Overlay Service (EGNOS), the Multi-functional Satellite Augmentation System (MSAS), the Global Positioning System (GPS) Aided Geo Augmented Navigation or GPS and Geo Augmented Navigation system (GAGAN), and/or the like. Thus, as used herein, a satellite positioning system may include any combination of one or more global and/or regional navigation satellites associated with such one or more satellite positioning systems.

[0069] In an aspect, SVs **112** may additionally or alternatively be part of one or more non-terrestrial networks (NTNs). In an NTN, an SV **112** is connected to an earth station (also referred to as a ground station, NTN gateway, or gateway), which in turn is connected to an element in a 5G network, such as a modified base station **102** (without a terrestrial antenna) or a network node in a 5GC. This element would in turn provide access to other elements in the 5G network and ultimately to entities external to the 5G network, such as Internet web servers and other user devices. In that way, a UE **104** may receive communication signals (e.g., signals **124**) from an SV **112** instead of, or in addition to, communication signals from a terrestrial base station **102**.

[0070] The wireless communications system **100** may further include one or more UEs, such as UE **190**, that connects indirectly to one or more communication networks via one or more device-to-device (D2D) peer-to-peer (P2P) links (referred to as “sidelinks”). In the example of FIG. **1**, UE **190** has a D2D P2P link **192** with one of the UEs **104** connected to one of the base stations **102** (e.g., through which UE **190** may indirectly obtain cellular connectivity) and a D2D P2P link **194** with WLAN STA **152** connected to the WLAN AP **150** (through which UE **190** may indirectly obtain WLAN-based Internet connectivity). In an example, the D2D P2P links **192** and **194** may be supported with any well-known D2D RAT, such as LTE Direct (LTE-D), WI-FI DIRECT®, BLUETOOTH®, and so on.

[0071] FIG. **2**A illustrates an example wireless network structure **200**. For example, a 5GC **210** (also referred to as a Next Generation Core (NGC)) can be viewed functionally as control plane (C-plane) functions **214** (e.g., UE registration, authentication, network access, gateway selection, etc.)

and user plane (U-plane) functions **212**, (e.g., UE gateway function, access to data networks, IP routing, etc.) which operate cooperatively to form the core network. User plane interface (NG-U) **213** and control plane interface (NG-C) **215** connect the gNB **222** to the 5GC **210** and specifically to the user plane functions **212** and control plane functions **214**, respectively. In an additional configuration, an ng-eNB **224** may also be connected to the 5GC **210** via NG-C **215** to the control plane functions **214** and NG-U **213** to user plane functions **212**. Further, ng-eNB **224** may directly communicate with gNB **222** via a backhaul connection **223**. In some configurations, a Next Generation RAN (NG-RAN) **220** may have one or more gNBs **222**, while other configurations include one or more of both ng-eNBs **224** and gNBs **222**. Either (or both) gNB **222** or ng-eNB **224** may communicate with one or more UEs **204** (e.g., any of the UEs described herein).

[0072] Another optional aspect may include a location server **230**, which may be in communication with the 5GC **210** to provide location assistance for UE(s) **204**. The location server **230** can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The location server **230** can be configured to support one or more location services for UEs **204** that can connect to the location server **230** via the core network, 5GC **210**, and/or via the Internet (not illustrated). Further, the location server **230** may be integrated into a component of the core network, or alternatively may be external to the core network (e.g., a third party server, such as an original equipment manufacturer (OEM) server or service server).

[0073] FIG. **2**B illustrates another example wireless network structure **240**. A 5GC **260** (which may correspond to 5GC **210** in FIG. **2**A) can be viewed functionally as control plane functions, provided by an access and mobility management function (AMF) **264**, and user plane functions, provided by a user plane function (UPF) **262**, which operate cooperatively to form the core network (i.e., 5GC **260**). The functions of the AMF **264** include registration management, connection management, reachability management, mobility management, lawful interception, transport for session management (SM) messages between one or more UEs **204** (e.g., any of the UEs described herein) and a session management function (SMF) **266**, transparent proxy services for routing SM messages, access authentication and access authorization, transport for short message service (SMS) messages between the UE **204** and the short message service function (SMSF) (not shown), and security anchor functionality (SEAF). The AMF **264** also interacts with an authentication server function (AUSF) (not shown) and the UE **204**, and receives the intermediate key that was established as a result of the UE **204** authentication process. In the case of authentication based on a UMTS (universal mobile telecommunications system) subscriber identity module (USIM), the AMF **264** retrieves the security material from the AUSF. The functions of the AMF **264** also include security context management (SCM). The SCM receives a key from the SEAF that it uses to derive access-network specific keys. The functionality of the AMF **264** also includes location services management for regulatory services, transport for location services messages between the UE **204** and a location management function (LMF) **270** (which acts as a location server **230**), transport for location services messages between the NG-RAN **220** and the LMF **270**, evolved packet system (EPS) bearer identifier allocation for interworking with the EPS, and UE **204** mobility event notification. In addition, the AMF **264** also supports functionalities for non-3GPP® (Third Generation Partnership Project) access networks.

[0074] Functions of the UPF **262** include acting as an anchor point for intra/inter-RAT mobility (when applicable), acting as an external protocol data unit (PDU) session point of interconnect to a data network (not shown), providing packet routing and forwarding, packet inspection, user plane policy rule enforcement (e.g., gating, redirection, traffic steering), lawful interception (user plane collection), traffic usage reporting, quality of service (QOS) handling for the user plane (e.g., uplink/downlink rate enforcement, reflective QoS marking in the downlink), uplink traffic verification (service data flow (SDF) to QoS flow mapping), transport level packet marking in the uplink and downlink, downlink packet buffering and downlink data notification triggering, and sending and forwarding of one or more "end markers" to the source RAN node. The UPF **262** may also support transfer of location services messages over a user plane between the UE **204** and a location server, such as an SLP **272**.

[0075] The functions of the SMF **266** include session management, UE Internet protocol (IP) address allocation and management, selection and control of user plane functions, configuration of traffic steering at the UPF **262** to route traffic to the proper destination, control of part of policy enforcement and QoS, and downlink data notification. The interface over which the SMF **266** communicates with the AMF **264** is referred to as the N11 interface.

[0076] Another optional aspect may include an LMF **270**, which may be in communication with the 5GC **260** to provide location assistance for UEs **204**. The LMF **270** can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The LMF **270** can be configured to support one or more location services for UEs **204** that can connect to the LMF **270** via the core network, 5GC **260**, and/or via the Internet (not illustrated). The SLP **272** may support similar functions to the LMF **270**, but whereas the LMF **270** may communicate with the AMF **264**, NG-RAN **220**, and UEs **204** over a control plane (e.g., using interfaces and protocols intended to convey signaling messages and not voice or data), the SLP **272** may communicate with UEs **204** and external clients (e.g., third-party server **274**) over a user plane (e.g., using protocols intended to carry voice and/or data like the transmission control protocol (TCP) and/or IP).

[0077] Yet another optional aspect may include a third-party server **274**, which may be in communication with the LMF **270**, the SLP **272**, the 5GC **260** (e.g., via the AMF **264** and/or the UPF **262**), the NG-RAN **220**, and/or the UE **204** to obtain location information (e.g., a location estimate) for the UE **204**. As such, in some cases, the third-party server **274** may be referred to as a location services (LCS) client or an external client. The third-party server **274** can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server.

[0078] User plane interface **263** and control plane interface **265** connect the 5GC **260**, and specifically the UPF **262** and AMF **264**, respectively, to one or more gNBs **222** and/or ng-eNBs **224** in the NG-RAN **220**. The interface between gNB(s) **222** and/or ng-cNB(s) **224** and the AMF **264** is referred to as the "N2" interface, and the interface between gNB(s) **222** and/or ng-eNB(s) **224** and the UPF **262** is referred to as the "N3" interface. The gNB(s) **222** and/or ng-eNB(s) **224** of the NG-RAN **220** may communicate directly with each other via backhaul connections **223**, referred to as the "Xn-C" interface. One or more of gNBs **222** and/or ng-eNBs **224** may communicate with one or more UEs **204** over a wireless interface, referred to as the "Uu" interface.

[0079] The functionality of a gNB **222** may be divided between a gNB central unit (gNB-CU) **226**, one or more gNB distributed units (gNB-DUs) **228**, and one or more gNB radio units (gNB-RUs) **229**. A gNB-CU **226** is a logical node that includes the base station functions of transferring user data, mobility control, radio access network sharing, positioning, session management, and the like, except for those functions allocated exclusively to the gNB-DU(s) **228**. More specifically, the gNB-CU **226** generally host the radio resource control (RRC), service data adaptation protocol (SDAP), and packet data convergence protocol (PDCP) protocols of the gNB **222**. A gNB-DU **228** is a logical node that generally hosts the radio link control (RLC) and medium access control (MAC) layer of the gNB **222**. Its operation is controlled by the gNB-CU **226**. One gNB-DU **228** can support one or more cells, and one cell is supported by only one gNB-DU **228**. The interface **232** between the gNB-CU **226** and the one or more gNB-DUs **228** is referred to as the "F1" interface. The physical (PHY) layer functionality of a gNB **222** is generally hosted by one or more standalone gNB-RUs **229** that perform functions such as power amplification and signal transmission/reception. The interface between a gNB-DU **228** and a gNB-RU **229** is referred to as the "Fx" interface. Thus, a UE **204** communicates with the gNB-CU **226** via the RRC, SDAP, and PDCP layers, with a gNB-DU **228** via the RLC and MAC layers, and with a gNB-RU **229** via the PHY layer.

[0080] FIGS. **3**A, **3**B, and **3**C illustrate several example components (represented by corresponding blocks) that may be incorporated into a UE **302** (which may correspond to any of the UEs described herein), a base station **304** (which may correspond to any of the base stations described herein), and a network entity **306** (which may correspond to or embody any of the network functions described herein, including the location server **230** and the LMF **270**, or alternatively may be independent from the NG-RAN **220** and/or 5GC **210**/**260** infrastructure depicted in FIGS. **2**A and **2**B, such as a private network) to support the operations described herein. It will be appreciated that these components may be implemented in different types of apparatuses in different implementations (e.g., in an ASIC, in a system-on-chip (SoC), etc.). The illustrated components may also be incorporated into other apparatuses in a communication system. For example, other apparatuses in a system may include components similar to those described to provide similar functionality. Also, a given apparatus may contain one or more of the components. For example, an apparatus may include multiple transceiver components that enable the apparatus to operate on multiple carriers and/or communicate via different technologies.

[0081] The UE **302** and the base station **304** each include one or more wireless wide area network (WWAN) transceivers **310** and **350**, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means for tuning, means for refraining from transmitting, etc.) via one or more wireless communication networks (not shown), such as an NR network, an LTE network, a GSM network, and/or the like. The WWAN transceivers **310** and **350** may each be connected to one or more antennas **316** and **356**, respectively, for communicating with other network nodes, such as other UEs, access points, base stations (e.g., eNBs, gNBs), etc., via at least one designated RAT (e.g., NR, LTE, GSM, etc.) over a wireless communication medium of interest (e.g., some set of time/frequency resources in a particular frequency spectrum). The WWAN transceivers **310** and **350** may be variously configured for transmitting and encoding signals **318** and **358** (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals **318** and **358** (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the WWAN transceivers **310** and **350** include one or more transmitters **314** and **354**, respectively, for transmitting and encoding signals **318** and **358**, respectively, and one or more receivers **312** and **352**, respectively, for receiving and decoding signals **318** and **358**, respectively.

[0082] The UE **302** and the base station **304** each also include, at least in some cases, one or more short-range wireless transceivers **320** and **360**, respectively. The short-range wireless transceivers **320** and **360** may be connected to one or more antennas **326** and **366**, respectively, and provide means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means for tuning, means for refraining from transmitting, etc.) with other network nodes, such as other UEs, access points, base stations, etc., via at least one designated RAT (e.g., Wi-Fi, LTE Direct, BLUETOOTH®, ZIGBEE®, Z-WAVE®, PC5, dedicated short-range communications (DSRC), wireless access for vehicular environments (WAVE), near-field communication (NFC), ultra-wideband (UWB), etc.) over a wireless communication medium of interest. The short-range wireless transceivers **320** and **360** may be variously configured for transmitting and encoding signals **328** and **368** (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals **328** and **368** (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the short-range wireless transceivers **320** and **360** include one or more transmitters **324** and **364**, respectively, for transmitting and encoding signals **328** and **368**, respectively, and one or more receivers **322** and **362**, respectively, for receiving and decoding signals **328** and **368**, respectively.

As specific examples, the short-range wireless transceivers **320** and **360** may be Wi-Fi transceivers, BLUETOOTH® transceivers, ZIGBEE® and/or Z-WAVE® transceivers, NFC transceivers, UWB transceivers, or vehicle-to-vehicle (V2V) and/or vehicle-to-everything (V2X) transceivers.

[0083] The UE **302** and the base station **304** also include, at least in some cases, satellite signal interfaces **330** and **370**, which each include one or more satellite signal receivers **332** and **372**, respectively, and may optionally include one or more satellite signal transmitters **334** and **374**, respectively. In some cases, the base station **304** may be a terrestrial base station that may communicate with space vehicles (e.g., space vehicles **112**) via the satellite signal interface **370**. In other cases, the base station **304** may be a space vehicle (or other non-terrestrial entity) that uses the satellite signal interface **370** to communicate with terrestrial networks and/or other space vehicles.

[0084] The satellite signal receivers **332** and **372** may be connected to one or more antennas **336** and **376**, respectively, and may provide means for receiving and/or measuring satellite positioning/communication signals **338** and **378**, respectively. Where the satellite signal receiver(s) **332** and **372** are satellite positioning system receivers, the satellite positioning/communication signals **338** and **378** may be global positioning system (GPS) signals, global navigation satellite system (GLONASS) signals, Galileo signals, Beidou signals, Indian Regional Navigation Satellite System (NAVIC), Quasi-Zenith Satellite System (QZSS) signals, etc. Where the satellite signal receiver(s) **332** and **372** are non-terrestrial network (NTN) receivers, the satellite positioning/communication signals **338** and **378** may be communication signals (e.g., carrying control and/or user data) originating from a 5G network. The satellite signal receiver(s) **332** and **372** may comprise any suitable hardware and/or software for receiving and processing satellite positioning/communication signals **338** and **378**, respectively. The satellite signal receiver(s) **332** and **372** may request information and operations as appropriate from the other systems, and, at least in some cases, perform calculations to determine locations of the UE **302** and the base station **304**, respectively, using measurements obtained by any suitable satellite positioning system algorithm.

[0085] The optional satellite signal transmitter(s) **334** and **374**, when present, may be connected to the one or more antennas **336** and **376**, respectively, and may provide means for transmitting satellite positioning/communication signals **338** and **378**, respectively. Where the satellite signal transmitter(s) **374** are satellite positioning system transmitters, the satellite positioning/communication signals **378** may be GPS signals, GLONASS® signals, Galileo signals, Beidou signals, NAVIC, QZSS signals, etc. Where the satellite signal transmitter(s) **334** and **374** are NTN transmitters, the satellite positioning/communication signals **338** and **378** may be communication signals (e.g., carrying control and/or user data) originating from a 5G network. The satellite signal transmitter(s) **334** and **374** may comprise any suitable hardware and/or software for transmitting satellite positioning/communication signals **338** and **378**, respectively. The satellite signal transmitter(s) **334** and **374** may request information and operations as appropriate from the other systems.

[0086] The base station **304** and the network entity **306** each include one or more network transceivers **380** and **390**, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, etc.) with other network entities (e.g., other base stations **304**, other network entities **306**). For example, the base station **304** may employ the one or more network transceivers **380** to communicate with other base stations **304** or network entities **306** over one or more wired or wireless backhaul links. As another example, the network entity **306** may employ the one or more network transceivers **390** to communicate with one or more base station **304** over one or more wired or wireless backhaul links, or with other network entities **306** over one or more wired or wireless core network interfaces.

[0087] A transceiver may be configured to communicate over a wired or wireless link. A transceiver (whether a wired transceiver or a wireless transceiver) includes transmitter circuitry (e.g., transmitters **314**, **324**, **354**, **364**) and receiver circuitry (e.g., receivers **312**, **322**, **352**, **362**). A transceiver may be an integrated device (e.g., embodying transmitter circuitry and receiver circuitry in a single device) in some implementations, may comprise separate transmitter circuitry and separate receiver circuitry in some implementations, or may be embodied in other ways in other implementations. The transmitter circuitry and receiver circuitry of a wired transceiver (e.g., network transceivers **380** and **390** in some implementations) may be coupled to one or more wired network interface ports. Wireless transmitter circuitry (e.g., transmitters **314**, **324**, **354**, **364**) may include or be coupled to a plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such as an antenna array, that permits the respective apparatus (e.g., UE **302**, base station **304**) to perform transmit "beamforming," as described herein. Similarly, wireless receiver circuitry (e.g., receivers **312**, **322**, **352**, **362**) may include or be coupled to a plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such as an antenna array, that permits the respective apparatus (e.g., UE **302**, base station **304**) to perform receive beamforming, as described herein. In an aspect, the transmitter circuitry and receiver circuitry may share the same plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such that the respective apparatus can only receive or transmit at a given time, not both at the same time. A wireless transceiver (e.g., WWAN transceivers **310** and **350**, short-range wireless transceivers **320** and **360**) may also include a network listen module (NLM) or the like for performing various measurements.

[0088] As used herein, the various wireless transceivers (e.g., transceivers **310**, **320**, **350**, and **360**, and network transceivers **380** and **390** in some implementations) and wired transceivers (e.g., network transceivers **380** and **390** in some implementations) may generally be characterized as "a transceiver," "at least one transceiver," or "one or more transceivers." As such, whether a particular transceiver is a wired or wireless transceiver may be inferred from the type of communication performed. For example, backhaul communication between network devices or servers will generally relate to signaling via a wired transceiver, whereas wireless communication between a UE (e.g., UE **302**) and a base station (e.g., base station **304**) will generally relate to signaling via a wireless transceiver.

[0089] The UE **302**, the base station **304**, and the network entity **306** also include other components that may be used in conjunction with the operations as disclosed herein. The UE **302**, the base station **304**, and the network entity **306** include one or more processors **342**, **384**, and **394**, respectively, for providing functionality relating to, for example, wireless communication, and for providing other processing functionality. The processors **342**, **384**, and **394** may therefore provide means for processing, such as means for determining, means for calculating, means for receiving, means for transmitting, means for indicating, etc. In an aspect, the processors **342**, **384**, and **394** may include, for example, one or more general purpose processors, multi-core processors, central processing units (CPUs), ASICs, digital signal processors (DSPs), field programmable gate arrays (FPGAs), other programmable logic devices or processing circuitry, or various combinations thereof.

[0090] The UE **302**, the base station **304**, and the network entity **306** include memory circuitry implementing memories **340**, **386**, and **396** (e.g., each including a memory device), respectively, for maintaining information (e.g., information indicative of reserved resources, thresholds, parameters, and so on). The memories **340**, **386**, and **396** may therefore provide means for storing, means for retrieving, means for maintaining, etc. In some cases, the UE **302**, the base station **304**, and the network entity **306** may include sensing component **348**, **388**, and **398**, respectively. The sensing component **348**, **388**, and **398** may be hardware circuits that are part of or coupled to the processors **342**, **384**, and **394**, respectively, that, when executed, cause the UE **302**, the base station **304**, and the network entity **306** to perform the functionality described herein. In other aspects, the sensing component **348**, **388**, and **398** may be external to the processors **342**, **384**, and **394** (e.g., part of a modem processing system, integrated with another processing system, etc.). Alternatively, the sensing component **348**, **388**, and **398** may be memory modules stored in the memories **340**, **386**, and **396**, respectively, that, when executed by the processors **342**, **384**, and **394** (or a modem processing system, another processing system, etc.), cause the UE **302**, the base station **304**, and the network entity **306** to perform the functionality described herein. FIG. **3**A illustrates possible locations of the sensing component **348**, which may be, for example, part of the one or more WWAN transceivers **310**, the memory **340**, the one or more processors **342**, or any combination thereof, or may be a standalone component. FIG. **3**B illustrates possible locations of the sensing component **388**, which may be, for example, part of the one or more WWAN transceivers **350**, the memory **386**, the one or more processors **384**, or any combination thereof, or may be a standalone component. FIG. **3**C illustrates possible locations of the sensing component **398**, which may be, for example, part of the one or more network transceivers **390**, the memory **396**, the one or more processors **394**, or any combination thereof, or may be a standalone component.

[0091] The UE **302** may include one or more sensors **344** coupled to the one or more processors **342** to provide means for sensing or detecting movement and/or orientation information that is independent of motion data derived from signals received by the one or more WWAN transceivers **310**, the one or more short-range wireless transceivers **320**, and/or the satellite signal interface **330**. By way of example, the sensor(s) **344** may include an accelerometer (e.g., a micro-electrical mechanical systems (MEMS) device), a gyroscope, a geomagnetic sensor (e.g., a compass), an altimeter (e.g., a barometric pressure altimeter), and/or any other type of movement detection sensor. Moreover, the sensor(s) **344** may include a plurality of different types of devices and combine their outputs in order to provide motion information. For example, the sensor(s) **344** may use a combination of a multi-axis accelerometer and orientation sensors to provide the ability to compute positions in two-dimensional (2D) and/or three-dimensional (3D) coordinate systems.

[0092] In addition, the UE **302** includes a user interface **346** providing means for providing indications (e.g., audible and/or visual indications) to a user and/or for receiving user input (e.g., upon user actuation of a sensing device such a keypad, a touch screen, a microphone, and so on). Although not shown, the base station **304** and the network entity **306** may also include user interfaces.

[0093] Referring to the one or more processors **384** in more detail, in the downlink, IP packets from the network entity **306** may be provided to the processor **384**. The one or more processors **384** may implement functionality for an RRC layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The one or more processors **384** may provide RRC layer functionality associated with broadcasting of system information (e.g., master information block (MIB), system information blocks (SIBs)), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter-RAT mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer PDUs, error correction through automatic repeat request (ARQ), concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, scheduling information reporting, error correction, priority handling, and logical channel prioritization.

[0094] The transmitter **354** and the receiver **352** may implement Layer-1 (L1) functionality associated with various signal processing functions. Layer-1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The transmitter **354** handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an orthogonal frequency division multiplexing (OFDM) subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an inverse fast Fourier transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM symbol stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE **302**. Each spatial stream may then be provided to one or more different antennas **356**. The transmitter **354** may modulate an RF carrier with a respective spatial stream for transmission.

[0095] At the UE **302**, the receiver **312** receives a signal through its respective antenna(s) **316**. The receiver **312** recovers information modulated onto an RF carrier and provides the information to the one or more processors **342**. The transmitter **314** and the receiver **312** implement Layer-1 functionality associated with various signal processing functions. The receiver **312** may perform spatial processing on the information to recover any spatial streams destined for the UE **302**. If multiple spatial streams are destined for the UE **302**, they may be combined by the receiver **312** into a single OFDM symbol stream. The receiver **312** then converts the OFDM symbol stream from the time-domain to the frequency domain using a fast Fourier transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station **304**. These soft decisions may be based on channel estimates computed by a channel estimator. The soft decisions are then decoded and de-interleaved to recover the data and control signals that were originally transmitted by the base station **304** on the physical channel. The data and control signals are then provided to the one or more processors **342**, which implements Layer-3 (L3) and Layer-2 (L2) functionality.

[0096] In the downlink, the one or more processors **342** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets from the core network. The one or more processors **342** are also responsible for error detection.

[0097] Similar to the functionality described in connection with the downlink transmission by the base station **304**, the one or more processors **342** provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through hybrid automatic repeat request (HARQ), priority handling, and logical channel prioritization.

[0098] Channel estimates derived by the channel estimator from a reference signal or feedback transmitted by the base station **304** may be used by the transmitter **314** to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the transmitter **314** may be provided to different antenna(s) **316**. The transmitter **314** may modulate an RF carrier with a respective spatial stream for transmission.

[0099] The uplink transmission is processed at the base station **304** in a manner similar to that described in connection with the receiver function at the UE **302**. The receiver **352** receives a signal through its respective antenna(s) **356**. The receiver **352** recovers information modulated onto an RF carrier and provides the information to the one or more processors **384**.

[0100] In the uplink, the one or more processors **384** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets from the UE **302**. IP packets from the one or more processors **384** may be provided to the core network. The one or more processors **384** are also responsible for error detection.

[0101] For convenience, the UE **302**, the base station **304**, and/or the network entity **306** are shown in FIGS. **3**A, **3**B, and **3**C as including various components that may be configured according to the various examples described herein. It will be appreciated, however, that the illustrated components may have different functionality in different designs. In particular, various components in FIGS. **3**A to **3**C are optional in alternative configurations and the various aspects include configurations that may vary due to design choice, costs, use of the device, or other considerations. For example, in case of FIG. **3**A, a particular implementation of UE **302** may omit the WWAN transceiver(s) **310** (e.g., a wearable device or tablet computer or personal computer (PC) or laptop may have Wi-Fi and/or BLUETOOTH® capability without cellular capability), or may omit the short-range wireless transceiver(s) **320** (e.g., cellular-only, etc.), or may omit the satellite signal interface **330**, or may omit the sensor(s) **344**, and so on. In another example, in case of FIG. **3**B, a particular implementation of the base station **304** may omit the WWAN transceiver(s) **350** (e.g., a Wi-Fi "hotspot" access point without cellular capability), or may omit the short-range wireless transceiver(s) **360** (e.g., cellular-only, etc.), or may omit the satellite signal interface **370**, and so on. For brevity, illustration of the various alternative configurations is not provided herein, but would be readily understandable to one skilled in the art.

[0102] The various components of the UE **302**, the base station **304**, and the network entity **306** may be communicatively coupled to each other over data buses **308**, **382**, and **392**, respectively. In an aspect, the data buses **308**, **382**, and **392** may form, or be part of, a communication interface of the UE **302**, the base station **304**, and the network entity **306**, respectively. For example, where different logical entities are embodied in the same device (e.g., gNB and location server functionality incorporated into the same base station **304**), the data buses **308**, **382**, and **392** may provide communication between them.

[0103] The components of FIGS. **3**A, **3**B, and **3**C may be implemented in various ways. In some implementations, the components of FIGS. **3**A, **3**B, and **3**C may be implemented in one or more circuits such as, for example, one or more processors and/or one or more ASICs (which may include one or more processors). Here, each circuit may use and/or incorporate at least one memory component for storing information or executable code used by the circuit to provide this functionality. For example, some or all of the functionality represented by blocks **310** to **346** may be implemented by processor and memory component(s) of the UE **302** (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Similarly, some or all of the functionality represented by blocks **350** to **388** may be implemented by processor and memory component(s) of the base station **304** (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Also, some or all of the functionality represented by blocks **390** to **398** may be implemented by processor and memory component(s) of the network entity **306** (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). For simplicity, various operations, acts, and/or functions are described herein as being performed "by a UE," "by a base station," "by a network entity," etc. However, as will be appreciated, such operations, acts, and/or functions may actually be performed by specific components or combinations of components of the UE **302**, base station **304**, network entity **306**, etc., such as the processors **342**, **384**, **394**, the transceivers **310**, **320**, **350**, and **360**, the memories **340**, **386**, and **396**, the sensing component **348**, **388**, and **398**, etc.

[0104] In some designs, the network entity **306** may be implemented as a core network component. In other designs, the network entity **306** may be distinct from a network operator or operation of the cellular network infrastructure (e.g., NG RAN **220** and/or 5GC **210**/**260**). For example, the network entity **306** may be a component of a private network that may be configured to communicate with the UE **302** via the base station **304** or independently from the base station **304** (e.g., over a non-cellular communication link, such as Wi-Fi).

[0105] NR supports a number of cellular network-based positioning technologies, including downlink-based, uplink-based, and downlink-and-uplink-based positioning methods. Downlink-based positioning methods include observed time difference of arrival (OTDOA) in LTE, downlink time difference of arrival (DL-TDOA) in NR, and downlink angle-of-departure (DL-AoD) in NR. FIG. **4** illustrates examples of various positioning methods, according to aspects of the disclosure. In an OTDOA or DL-TDOA positioning procedure, illustrated by scenario **410**, a UE measures the differences between the times of arrival (ToAs) of reference signals (e.g., positioning reference signals (PRS)) received from pairs of base stations, referred to as reference signal time difference (RSTD) or time difference of arrival (TDOA) measurements, and reports them to a positioning entity. More specifically, the UE receives the identifiers (IDs) of a reference base station (e.g., a serving base station) and multiple non-reference base stations in assistance data. The UE then measures the RSTD between the reference base station and each of the non-reference base stations. Based on the known locations of the involved base stations and the RSTD measurements, the positioning entity (e.g., the UE for UE-based positioning or a location server for UE-assisted positioning) can estimate the UE's location.

[0106] For DL-AoD positioning, illustrated by scenario **420**, the positioning entity uses a measurement report from the UE of received signal strength measurements of multiple downlink transmit beams to determine the angle(s) between the UE and the transmitting base station(s). The positioning entity can then estimate the location of the UE based on the determined angle(s) and the known location(s) of the transmitting base station(s).

[0107] Uplink-based positioning methods include uplink time difference of arrival (UL-TDOA) and uplink angle-of-arrival (UL-AoA). UL-TDOA is similar to DL-TDOA, but is based on uplink reference signals (e.g., sounding reference signals (SRS)) transmitted by the UE to multiple base stations. Specifically, a UE transmits one or more uplink reference signals that are measured by a reference base station and a plurality of non-reference base stations. Each base station then reports the reception time (referred to as the relative time of arrival (RTOA)) of the reference signal(s) to a positioning entity (e.g., a location server) that knows the locations and relative timing of the involved base stations. Based on the reception-to-reception (Rx-Rx) time difference between the reported RTOA of the reference base station and the reported RTOA of each non-reference base station, the known locations of the base stations, and their known timing offsets, the positioning entity can estimate the location of the UE using TDOA.

[0108] For UL-AoA positioning, one or more base stations measure the received signal strength of one or more uplink reference signals (e.g., SRS) received from a UE on one or more uplink receive beams. The positioning entity uses the signal strength measurements and the angle(s) of the receive beam(s) to determine the angle(s) between the UE and the base station(s). Based on the determined angle(s) and the known location(s) of the base station(s), the positioning entity can then estimate the location of the UE.

[0109] Downlink-and-uplink-based positioning methods include enhanced cell-ID (E-CID) positioning and multi-round-trip-time (RTT) positioning (also referred to as “multi-cell RTT” and “multi-RTT”). In an RTT procedure, a first entity (e.g., a base station or a UE) transmits a first RTT-related signal (e.g., a PRS or SRS) to a second entity (e.g., a UE or base station), which transmits a second RTT-related signal (e.g., an SRS or PRS) back to the first entity. Each entity measures the time difference between the time of arrival (ToA) of the received RTT-related signal and the transmission time of the transmitted RTT-related signal. This time difference is referred to as a reception-to-transmission (Rx-Tx) time difference. The Rx-Tx time difference measurement may be made, or may be adjusted, to include only a time difference between nearest slot boundaries

for the received and transmitted signals. Both entities may then send their Rx-Tx time difference measurement to a location server (e.g., an LMF **270**), which calculates the round trip propagation time (i.e., RTT) between the two entities from the two Rx-Tx time difference measurements (e.g., as the sum of the two Rx-Tx time difference measurements). Alternatively, one entity may send its Rx-Tx time difference measurement to the other entity, which then calculates the RTT. The distance between the two entities can be determined from the RTT and the known signal speed (e.g., the speed of light). For multi-RTT positioning, illustrated by scenario **430**, a first entity (e.g., a UE or base station) performs an RTT positioning procedure with multiple second entities (e.g., multiple base stations or UEs) to enable the location of the first entity to be determined (e.g., using multilateration) based on distances to, and the known locations of, the second entities. RTT and multi-RTT methods can be combined with other positioning techniques, such as UL-AoA and DL-AoD, to improve location accuracy, as illustrated by scenario **440**.
[0110] The E-CID positioning method is based on radio resource management (RRM) measurements. In E-CID, the UE reports the serving cell ID, the timing advance (TA), and the identifiers, estimated timing, and signal strength of detected neighbor base stations. The location of the UE is then estimated based on this information and the known locations of the base station(s).
[0111] To assist positioning operations, a location server (e.g., location server **230**, LMF **270**, SLP **272**) may provide assistance data to the UE. For example, the assistance data may include identifiers of the base stations (or the cells/TRPs of the base stations) from which to measure reference signals, the reference signal configuration parameters (e.g., the number of consecutive slots including PRS, periodicity of the consecutive slots including PRS, muting sequence, frequency hopping sequence, reference signal identifier, reference signal bandwidth, etc.), and/or other parameters applicable to the particular positioning method. Alternatively, the assistance data may originate directly from the base stations themselves (e.g., in periodically broadcasted overhead messages, etc.). In some cases, the UE may be able to detect neighbor network nodes itself without the use of assistance data.
[0112] In the case of an OTDOA or DL-TDOA positioning procedure, the assistance data may further include an expected RSTD value and an associated uncertainty, or search window, around the expected RSTD. In some cases, the value range of the expected RSTD may be +/−500 microseconds (μs). In some cases, when any of the resources used for the positioning measurement are in FR1, the value range for the uncertainty of the expected RSTD may be +/−32 μs. In other cases, when all of the resources used for the positioning measurement(s) are in FR2, the value range for the uncertainty of the expected RSTD may be +/−8 μs.
[0113] A location estimate may be referred to by other names, such as a position estimate, location, position, position fix, fix, or the like. A location estimate may be geodetic and comprise coordinates (e.g., latitude, longitude, and possibly altitude) or may be civic and comprise a street address, postal address, or some other verbal description of a location. A location estimate may further be defined relative to some other known location or defined in absolute terms (e.g., using latitude, longitude, and possibly altitude). A location estimate may include an expected error or uncertainty (e.g., by including an area or volume within which the location is expected to be included with some specified or default level of confidence).
[0114] Wireless communication signals (e.g., radio frequency (RF) signals configured to carry orthogonal frequency division multiplexing (OFDM) symbols in accordance with a wireless communications standard, such as LTE, NR, etc.) transmitted between a UE and a base station can be used for environment sensing (also referred to as “RF sensing” or “radar”). Using wireless communication signals for environment sensing can be regarded as consumer-level radar with advanced detection capabilities that enable, among other things, touchless/device-free interaction with a device/system. The wireless communication signals may be cellular communication signals, such as LTE or NR signals, WLAN signals, such as Wi-Fi signals, etc. As a particular example, the wireless communication signals may be an OFDM waveform as utilized in LTE and NR. High-

frequency communication signals, such as millimeter wave (mmW) RF signals, are especially beneficial to use as sensing signals because the higher frequency provides, at least, more accurate range (distance) detection.

[0115] Possible use cases of RF sensing include health monitoring use cases, such as heartbeat detection, respiration rate monitoring, and the like, gesture recognition use cases, such as human activity recognition, keystroke detection, sign language recognition, and the like, contextual information acquisition use cases, such as location detection/tracking, direction finding, range estimation, and the like, and automotive sensing use cases, such as smart cruise control, collision avoidance, and the like.

[0116] There are different types of sensing, including monostatic sensing (also referred to as “active sensing”) and bistatic sensing (also referred to as “passive sensing”). FIGS. **5**A and **5**B illustrate these different types of sensing. Specifically, FIG. **5**A is a diagram **500** illustrating a monostatic sensing scenario and FIG. **5**B is a diagram **530** illustrating a bistatic sensing scenario. In FIG. **5**A, the transmitter (Tx) and receiver (Rx) are co-located in the same sensing device **504** (e.g., a UE). The sensing device **504** transmits one or more RF sensing signals **534** (e.g., uplink or sidelink positioning reference signals (PRS) where the sensing device **504** is a UE), and some of the RF sensing signals **534** reflect off a target object **506**. The sensing device **504** can measure various properties (e.g., times of arrival (ToAs), angles of arrival (AoAs), phase shift, etc.) of the reflections **536** of the RF sensing signals **534** to determine characteristics of the target object **506** (e.g., size, shape, speed, motion state, etc.).

[0117] In FIG. **5**B, the transmitter (Tx) and receiver (Rx) are not co-located, that is, they are separate devices (e.g., a UE and a base station). Note that while FIG. **5**B illustrates using a downlink RF signal as the RF sensing signal **532**, uplink RF signals or sidelink RF signals can also be used as RF sensing signals **532**. In a downlink scenario, as shown, the transmitter is a base station and the receiver is a UE, whereas in an uplink scenario, the transmitter is a UE and the receiver is a base station.

[0118] Referring to FIG. **5**B in greater detail, the transmitter device **502** transmits RF sensing signals **532** and **534** (e.g., positioning reference signals (PRS)) to the sensing device **504**, but some of the RF sensing signals **534** reflect off a target object **506**. The sensing device **504** (also referred to as the “sensing device”) can measure the times of arrival (ToAs) of the RF sensing signals **532** received directly from the transmitter device and the ToAs of the reflections **536** of the RF sensing signals **534** reflected from the target object **506**.

[0119] More specifically, as described above, a transmitter device (e.g., a base station) may transmit a single RF signal or multiple RF signals to a sensing device (e.g., a UE). However, the receiver may receive multiple RF signals corresponding to each transmitted RF signal due to the propagation characteristics of RF signals through multipath channels. Each path may be associated with a cluster of one or more channel taps. Generally, the time at which the receiver detects the first cluster of channel taps is considered the ToA of the RF signal on the line-of-site (LOS) path (i.e., the shortest path between the transmitter and the receiver). Later clusters of channel taps are considered to have reflected off objects between the transmitter and the receiver and therefore to have followed non-LOS (NLOS) paths between the transmitter and the receiver.

[0120] Thus, referring back to FIG. 5B, the RF sensing signals **532** followed the LOS path between the transmitter device **502** and the sensing device **504**, and the RF sensing signals **534** followed an NLOS path between the transmitter device **502** and the sensing device **504** due to reflecting off the target object **506**. The transmitter device **502** may have transmitted multiple RF sensing signals **532**, **534**, some of which followed the LOS path and others of which followed the NLOS path. Alternatively, the transmitter device **502** may have transmitted a single RF sensing signal in a broad enough beam that a portion of the RF sensing signal followed the LOS path (RF sensing signal **532**) and a portion of the RF sensing signal followed the NLOS path (RF sensing signal **534**).

[0121] Based on the ToA of the LOS path, the ToA of the NLOS path, and the speed of light, the sensing device **504** can determine the distance to the target object(s). For example, the sensing device **504** can calculate the distance to the target object as the difference between the ToA of the LOS path and the ToA of the NLOS path multiplied by the speed of light. In addition, if the sensing device **504** is capable of receive beamforming, the sensing device **504** may be able to determine the general direction to a target object as the direction (angle) of the receive beam on which the RF sensing signal following the NLOS path was received. That is, the sensing device **504** may determine the direction to the target object as the angle of arrival (AoA) of the RF sensing signal, which is the angle of the receive beam used to receive the RF sensing signal. The sensing device **504** may then optionally report this information to the transmitter device **502**, its serving base station, an application server associated with the core network, an external client, a third-party application, or some other sensing entity. Alternatively, the sensing device **504** may report the ToA measurements to the transmitter device **502**, or other sensing entity (e.g., if the sensing device **504** does not have the processing capability to perform the calculations itself), and the transmitter device **502** may determine the distance and, optionally, the direction to the target object **506**.

[0122] Note that if the RF sensing signals are uplink RF signals transmitted by a UE to a base station, the base station would perform object detection based on the uplink RF signals just like the UE does based on the downlink RF signals.

[0123] Like conventional radar, wireless communication-based sensing signals can be used to estimate the range (distance), velocity (Doppler), and angle (AoA) of a target object. However, the performance (e.g., resolution and maximum values of range, velocity, and angle) may depend on the design of the reference signal.

[0124] Various frame structures may be used to support downlink and uplink transmissions between network nodes (e.g., base stations and UEs). FIG. **6** is a diagram **600** illustrating an example frame structure, according to aspects of the disclosure. The frame structure may be a downlink or uplink frame structure. Other wireless communications technologies may have different frame structures and/or different channels.

[0125] LTE, and in some cases NR, utilizes orthogonal frequency-division multiplexing (OFDM) on the downlink and single-carrier frequency division multiplexing (SC-FDM) on the uplink. Unlike LTE, however, NR has an option to use OFDM on the uplink as well. OFDM and SC-FDM partition the system bandwidth into multiple (K) orthogonal subcarriers, which are also commonly referred to as tones, bins, etc. Each subcarrier may be modulated with data. In general, modulation symbols are sent in the frequency domain with OFDM and in the time domain with SC-FDM. The spacing between adjacent subcarriers may be fixed, and the total number of subcarriers (K) may be dependent on the system bandwidth. For example, the spacing of the subcarriers may be 15 kilohertz (kHz) and the minimum resource allocation (resource block) may be 12 subcarriers (or 180 kHz). Consequently, the nominal fast Fourier transform (FFT) size may be equal to 128, 256, 512, 1024, or 2048 for system bandwidth of 1.25, 2.5, 5, 10, or 20 megahertz (MHz), respectively. The system bandwidth may also be partitioned into subbands. For example, a subband may cover 1.08 MHz (i.e., **6** resource blocks), and there may be 1, 2, 4, 8, or 16 subbands for system bandwidth of 1.25, 2.5, 5, 10, or 20 MHZ, respectively.

[0126] LTE supports a single numerology (subcarrier spacing (SCS), symbol length, etc.). In contrast, NR may support multiple numerologies (μ), for example, subcarrier spacings of 15 kHz (μ=0), 30 kHz (μ=1), 60 kHz (μ=2), 120 kHz (μ=3), and 240 kHz (μ=4) or greater may be available. In each subcarrier spacing, there are 14 symbols per slot. For 15 kHz SCS (μ=0), there is one slot per subframe, 10 slots per frame, the slot duration is 1 millisecond (ms), the symbol duration is 66.7 microseconds (us), and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 50. For 30 kHz SCS (μ=1), there are two slots per subframe, 20 slots per frame, the slot duration is 0.5 ms, the symbol duration is 33.3 μs, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 100. For 60 kHz SCS (μ=2), there are four slots per subframe, 40 slots per frame, the slot duration is 0.25 ms, the symbol duration is 16.7 μs, and the

maximum nominal system bandwidth (in MHz) with a 4K FFT size is 200. For 120 kHz SCS (μ=3), there are eight slots per subframe, 80 slots per frame, the slot duration is 0.125 ms, the symbol duration is 8.33 μs, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 400. For 240 kHz SCS (μ=4), there are 16 slots per subframe, 160 slots per frame, the slot duration is 0.0625 ms, the symbol duration is 4.17 μs, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 800.
[0127] In the example of FIG. **6**, a numerology of 15 kHz is used. Thus, in the time domain, a 10 ms frame is divided into 10 equally sized subframes of 1 ms each, and each subframe includes one time slot. In FIG. **6**, time is represented horizontally (on the X axis) with time increasing from left to right, while frequency is represented vertically (on the Y axis) with frequency increasing (or decreasing) from bottom to top.
[0128] A resource grid may be used to represent time slots, each time slot including one or more time-concurrent resource blocks (RBs) (also referred to as physical RBs (PRBs)) in the frequency domain. The resource grid is further divided into multiple resource elements (REs). An RE may correspond to one symbol length in the time domain and one subcarrier in the frequency domain. In the numerology of FIG. **6**, for a normal cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and seven consecutive symbols in the time domain, for a total of 84 REs. For an extended cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and six consecutive symbols in the time domain, for a total of 72 REs. The number of bits carried by each RE depends on the modulation scheme.
[0129] Some of the REs may carry reference (pilot) signals (RS). The reference signals may include positioning reference signals (PRS), tracking reference signals (TRS), phase tracking reference signals (PTRS), cell-specific reference signals (CRS), channel state information reference signals (CSI-RS), demodulation reference signals (DMRS), primary synchronization signals (PSS), secondary synchronization signals (SSS), synchronization signal blocks (SSBs), sounding reference signals (SRS), etc., depending on whether the illustrated frame structure is used for uplink or downlink communication. FIG. **6** illustrates example locations of REs carrying a reference signal (labeled “R”).
[0130] In an aspect, the reference signal carried on the REs labeled “R” in FIG. **6** may be SRS. SRS transmitted by a UE may be used by a base station to obtain the channel state information (CSI) for the transmitting UE. CSI describes how an RF signal propagates from the UE to the base station and represents the combined effect of scattering, fading, and power decay with distance. The system uses the SRS for resource scheduling, link adaptation, massive MIMO, beam management, etc.
[0131] A collection of REs that are used for transmission of SRS is referred to as an “SRS resource,” and may be identified by the parameter “SRS-ResourceId.” The collection of resource elements can span multiple PRBs in the frequency domain and ‘N’ (e.g., one or more) consecutive symbol(s) within a slot in the time domain. In a given OFDM symbol, an SRS resource occupies one or more consecutive PRBs. An “SRS resource set” is a set of SRS resources used for the transmission of SRS signals, and is identified by an SRS resource set ID (“SRS-ResourceSetId”).
[0132] Generally, as noted above, a UE transmits SRS to enable the receiving base station (either the serving base station or a neighboring base station) to measure the channel quality (i.e., CSI) between the UE and the base station. However, SRS can also be specifically configured as uplink positioning reference signals for uplink-based positioning procedures, such as UL-TDOA, RTT, UL-AoA, etc. As used herein, the term “SRS” may refer to SRS configured for channel quality measurements or SRS configured for positioning purposes. The former may be referred to herein as “SRS-for-communication” and/or the latter may be referred to as “SRS-for-positioning” or “positioning SRS” when needed to distinguish the two types of SRS.
[0133] Several enhancements over the previous definition of SRS may be available for SRS-for-positioning (also referred to as “UL-PRS”), such as a new staggered pattern within an SRS resource

(except for single-symbol/comb-2), a new comb type for SRS, new sequences for SRS, a higher number of SRS resource sets per component carrier, and a higher number of SRS resources per component carrier. In addition, the parameters "SpatialRelationInfo" and "PathLossReference" are to be configured based on a downlink reference signal or SSB from a neighboring TRP. Further still, one SRS resource may be transmitted outside the active bandwidth part (BWP), and one SRS resource may span across multiple component carriers. Also, SRS may be configured in RRC connected state and only transmitted within an active BWP. Further, there may be no frequency hopping, no repetition factor, a single antenna port, and new lengths for SRS (e.g., 8 and 12 symbols). There also may be open-loop power control and not closed-loop power control, and comb-8 (i.e., an SRS transmitted every eighth subcarrier in the same symbol) may be used. Lastly, the UE may transmit through the same transmit beam from multiple SRS resources for UL-AoA. These features may be configured through RRC higher layer signaling (and potentially triggered or activated through a MAC control element (MAC-CE) or downlink control information (DCI)).
[0134] The following table illustrates transmission properties of SRS-for-positioning in relation to SRS-for-communication.
TABLE-US-00001 TABLE 1 Transmission Adjustments/Enhancements of SRS- properties similar to for-positioning over SRS-for- SRS-for-communication communication Same Zadoff-Chu Path-loss reference can be a PRS or an sequences SSB from a serving or neighboring cell (only open loop power control) SRS is configured and A multi-symbol SRS resource is always transmitted within staggered in frequency an active BWP Periodic, Semi-persistent, No frequency hopping Aperiodic Same timing advance Single-port as regular uplink channels Group or sequence If SRS-for-positioning collides with hopping is supported physical uplink shared channel (PUSCH), the SRS is dropped in the symbols where the collision occurs SRS can be configured in any OFDM symbol of a slot Spatial Relation Info can be a DL-PRS, an SSB or another SRS for positioning
[0135] The transmission of SRS resources within a given PRB has a particular comb size (also referred to as the "comb density"). A comb size 'N' represents the subcarrier spacing (or frequency/tone spacing) within each symbol of an SRS resource configuration. Specifically, for a comb size 'N,' SRS are transmitted in every Nth subcarrier of a symbol of a PRB. For example, for comb-4, for each symbol of the SRS resource configuration, REs corresponding to every fourth subcarrier (such as subcarriers 0, 4, 8) are used to transmit SRS of the SRS resource. In the example of FIG. **6**, the illustrated SRS is comb-4 over four symbols. That is, the locations of the shaded SRS REs indicate a comb-4 SRS resource configuration.
[0136] Currently, an SRS resource may span 1, 2, 4, 8, or 12 consecutive symbols within a slot with a comb size of comb-2, comb-4, or comb-8. The following table shows the frequency offsets from symbol to symbol for the SRS comb patterns that are currently supported.
TABLE-US-00002 TABLE 2 1 2 4 Comb Symbol Symbols Symbols 8 Symbols 12 Symbols 2 {0} {0, 1} {0, 1, 0, 1} NA NA 4 NA {0, 2} {0, 2, 1, 3} {0, 2, {0, 2, 1, 3, 0, 1, 3, 0, 2, 1, 3, 0, 2, 1, 3} 2, 1, 3} 8 NA NA {0, 4, 2, 6} {0, 4, {0, 4, 2, 6, 1, 2, 6, 1, 5, 3, 7, 0, 5, 3, 7} 4, 2, 6}
[0137] FIGS. **7**A to **7**C illustrate various comb patterns supported for SRS within a resource block. The illustrated comb patterns correspond to various comb patterns shown in Table 2 above. Each comb pattern is illustrated within a single resource block. In each resource block, frequency is represented vertically and time is represented horizontally. Each block of each resource block represents an RE, and each shaded block represents an RE carrying SRS. All of the REs carrying SRS within a resource block correspond to an SRS resource.
[0138] FIG. **7**A illustrates a comb pattern **710** for comb-2 with one symbol, a comb pattern **720** for comb-2 with two symbols, and a comb pattern **730** for comb-2 with four symbols. FIG. **7**B illustrates a comb pattern **740** for comb-4 with two symbols, a comb pattern **750** for comb-4 with four symbols, and a comb pattern **760** for comb-4 with eight symbols. The comb pattern for comb-4 with 12 symbols is not illustrated, but as shown in Table 2, it is similar to comb-4 with eight symbols but with four more symbols having a repeated pattern of offsets {0, 2, 1, 3}. FIG. **7**C

illustrates a comb pattern **770** for comb-8 with four symbols, a comb pattern **780** for comb-8 with eight symbols, and a comb pattern **790** for comb-8 with 12 symbols.

[0139] A frequency modulated continuous wave (FMCW) waveform is a signal where the frequency increases (up-chirp) or decreases (down-chirp) with time linearly. FIGS. **8** and **9** illustrate graphs **800** and **900**, respectively, of example FMCW waveforms, according to aspects of the disclosure.

[0140] FMCW waveforms provide certain technical advantages with respect to RF sensing operations (as described above with reference to FIGS. **5**A and **5**B). For example, FMCW waveforms may be used to perform sensing and to obtain a channel estimate using narrowband (NB) baseband processing. As another example, FMCW waveforms may be used to perform wideband (WB) sensing and to obtain a channel estimate using NB baseband processing. FMCW waveforms may provide a high performance to cost ratio (e.g., a low-speed analog to digital converter (ADC) could be used to sample the beat signal, e.g., from several GHz or 100s of MHz to 10s of MHz or even less than 10 MHZ). FMCW waveforms may also provide low peak-to-average power ratio (PAPR), facilitating low complexity full duplex sensing.

[0141] FIG. **10** illustrates an example FMCW system **1000**, according to aspects of the disclosure. In FIG. **10**, a received signal is mixed with the transmitted FMCW waveform and passed through a low pass filter (LPF) to generate a narrowband beat signal, which is input to an ADC. Each beat signal frequency fb maps to a specific target reflection.

[0142] FIG. **11** is a diagram **1100** illustrating the two types of cyclic prefix OFDM (CP-OFDM) compatible FMCW waveforms for analog transmission and/or sensing, according to aspects of the disclosure. In FIG. **11**, each block is the length of a cyclic prefix and a symbol. As shown in FIG. **11**, for a Type 1 CP-OFDM compatible FMCW waveform, there is a frequency jump between the cyclic prefix (CP) and the symbol. In contrast, a Type 2 CP-OFDM compatible FMCW waveform is analog-generated friendly since the gap can be used for voltage controlled oscillator (VCO) reset.

[0143] FIG. **12** illustrates a comparison of analog versus digital generation of a Type 2 FMCW, according to aspects of the disclosure. Specifically, graph **1200** illustrates an example analog-generated FMCW waveform and graph **1250** illustrates an example digital-generated FMCW waveform. As shown in graph **1250**, there is distortion error in the digital-generated FMCW waveform.

[0144] There are different types of CP-OFDM compatible FMCW waveform designs, including comb-based frequency division multiplexing (FDM) designs, such as zero-tail FMCW (as illustrated in FIGS. **11** and **12**) or piecewise linear chirp with a cyclic prefix structure. In these designs, within the OFDM symbol duration, the FMCW is transmitted repeatedly with a triangular chirp shape to facilitate chirp repetition using an analog VCO-based implementation (e.g., as illustrated in FIGS. **11** and **12**). The following table illustrates example FDM designs and the corresponding FMCW numerology.

TABLE-US-00003 TABLE 3 Example FDM Design FMCW Numerology Notes Zero CP based T.sub.OFDM = α Support any type of FDM (T.sub.CP + T.sub.FMCW) FMCW The gaps with CP duration enable RF tuning between FMCW instances CP compatible T.sub.CP = (β + δ)T.sub.FMCW or CP consists of (integer + FMCW T.sub.CP = δT.sub.FMCW fractional) or based FDM T.sub.OFDM = γT.sub.FMCW fractional number of chirps OFDM consists of integer number of chirps Fractional chirp duration δ for legacy numerology designs

[0145] In the above equations, the comb a or y resource elements (REs) are allocated to FMCW and other REs are allocated to OFDM. The FMCW and OFDM waveforms may therefore orthogonally coexist.

[0146] FIG. **13** illustrates the example FDM designs described in Table 3, according to aspects of the disclosure. Specifically, diagram **1300** illustrates an example zero CP based FDM waveform and diagram **1350** illustrates two example piecewise linear chirp with CP structure waveforms.

[0147] As described above, 5G NR has already adopted SRS for both communications and UE positioning with different “usage.” Even though SRS for positioning is enhanced for UE positioning, it has not yet been deployed in the commercial networks. As such, it has been proposed to reuse SRS-for-communication for UE positioning.

[0148] One design direction for 6G may be a unified reference signal across multiple use cases, such as communication channel estimation/sounding, UE positioning, and RF sensing. In that case, there may be a unified SRS for these multiple use cases. In the case of a unified SRS with multiple functionalities, the SRS would likely also be used for non-communications use cases, such as RF sensing.

[0149] Other challenges of the existing SRS that may be addressed by a unified SRS include coverage enhancement due to uplink transmission, positioning/RF sensing capacity, and improvements to UE implementation and power saving, especially for higher bandwidths (such as 1 GHz component carriers).

[0150] Accordingly, the present disclosure provides an enhanced SRS waveform to enable SRS to be used for multiple use cases. The enhanced SRS waveform may be configured as a CP-OFDM compatible FMCW waveform, which could support multiple types of FMCW for efficient uplink transmission and full duplex sensing. In legacy SRS, there is a frequency jump either between the cyclic prefix and the OFDM symbol, or within the OFDM symbol, and as such, it is not “friendly” for a VCO based implementation. However, the VCO based analog implementation is an important aspect to achieving the advantages of FMCW, due to the small baseband bandwidth to achieve large RF bandwidth transmission and/or reception, low complexity self-interference cancelation, and the like.

[0151] In some designs, a CP-OFDM compatible FMCW waveform could enable efficient multiplexing between SRS and physical uplink shared channel (PUSCH)/physical uplink control channel (PUCCH), such as comb-based multiplexing between SRS and PUSCH/PUCCH.

[0152] The configuration of an enhanced SRS would depend on the UE's capabilities, in particular, whether the UE supports a CP-OFDM compatible FMCW waveform. If the UE supports a CP-OFDM compatible FMCW waveform, the UE may indicate which type(s) of CP-OFDM compatible FMCW waveforms it supports, such as zero tail FMCW, piecewise linear chirp with cyclic prefix structure, or other CP-OFDM compatible chirp. In some designs, the network may schedule a group of UEs supporting the same type of CP-OFDM compatible FMCW waveform for multiplexing. UEs supporting different types of CP-OFDM compatible FMCW waveforms may not be multiplexed.

[0153] If the UE does not support CP-OFDM compatible FMCW waveforms, the network would only configure the legacy SRS for the UE. Note that the legacy SRS waveform could still be multiplexed with the enhanced SRS with the enhanced waveform. However, the legacy SRS waveform may not support ultra-wideband RF sensing and positioning.

[0154] The UE may further indicate whether it supports multiplexing between the enhanced SRS and PUSCH/PUCCH. Comb-based multiplexing between SRS and PUSCH/PUCCH may depend on the UE's capability to concurrently support FMCW transmission and OFDM transmission.

[0155] The enhanced SRS may still be staggered with multiple symbols for better coverage, and there may be multiple symbol transmission to extend the coverage. As such, a UE would need to support CP-OFDM compatible FMCW transmission with different comb offsets in different symbols (and report such a capability). The UE would also need to indicate whether it is capable of performing enhanced SRS transmission with different comb offsets back-to-back. If the UE cannot support back-to-back SRS transmission with different comb offsets, the following design may be considered. One or more gap symbols may be configured between SRS symbols and the gap symbol(s) may be used by other users and/or for other uplink channels (e.g., PUSCH, PUCCH).

[0156] FIG. **14** illustrates a comparison of back-to-back staggered SRS without gaps and staggered SRS with gaps, according to aspects of the disclosure. Specifically, diagram **1400** illustrates an example of back-to-back staggered SRS without gap symbols and diagram **1450** illustrates an example of staggered SRS with one gap symbols between SRS symbols.

[0157] Currently, SRS is configured and transmitted within an active BWP. Note that the BWP refers to the baseband bandwidth, not the RF bandwidth. More specifically, the baseband bandwidth is based on the UE's analog-to-digital converter (ADC) sampling rate and is what the UE hardware can process. The UE's capability for baseband processing depends on data communications use cases, and as such, its baseband bandwidth may be, for example, 10 MHz or 100 MHZ In contrast, the RF bandwidth is the bandwidth of what is transmitted over the air and is usually a much larger bandwidth than the baseband BWP (e.g., 1 GHZ). The UE can only sample a portion (the baseband BWP) of the RF bandwidth. In NR, all the BWP configurations refer is the baseband bandwidth.

[0158] Some UEs may not support a wideband baseband, however, they may still need to support ultra-wideband 6G SRS transmission. That is, the BWP supported by/configured to the UE may be much smaller than that of the enhanced SRS. However, as discussed above, the enhanced SRS waveform is a CP-OFDM compatible FMCW waveform, which could facilitate a VCO based analog implementation. As such, it can enable a UE's X GHz SRS transmission with a Y MHz baseband bandwidth, where Y is decided by the communication requirements.

[0159] As a first option, a new type of uplink BWP for the enhanced SRS transmission may be defined. In this case, the new type of uplink BWP may be the RF BWP, or other larger BWP, instead of the narrower legacy baseband BWP. In some designs, the new type of larger uplink BWP may only be applied to the enhanced SRS in the uplink. As a second option, there may be measurement gap-based SRS transmission. In this case, the network may configure a gap for the wideband SRS transmission. However, the overhead for this option might be higher.

[0160] In some designs, a UE may need to report several related capabilities, specifically, whether the UE can concurrently support the legacy BWP and the new type of BWP and the UE's BWP switching capability. If the UE can concurrently support the legacy BWP and the new type of BWP, the network can decide whether to schedule SRS and PUSCH/PUCCH within the same time window. The UE may also indicate whether it can support concurrent SRS and PUSCH/PUCCH transmission and whether the SRS transmission is with a different hardware chain.

[0161] Referring to the UE's BWP switching capability, this capability indicates, for example, how much time is needed for the switch between the legacy BWP and the new type of BWP. FIG. **15** is a diagram **1500** illustrating an example of different BWP switching latencies, according to aspects of the disclosure. As shown in FIG. **15**, there is a relatively shorter switching time between two legacy BWPs (labeled “BWP1” and “BWP2”) and the new type of BWP (labeled “BWP for 6G SRS”).

[0162] In some designs, to reduce the SRS overhead, different SRS could occupy the same resource(s), but they could be orthogonal via Doppler-division multiplexing (DDM). FIGS. **16**A and **16**B illustrate various concepts of DDM, according to aspects of the disclosure. Specifically, diagram **1600** illustrates an example of slow time phase coding for M.sub.t SRS antenna ports. The SRS ports may be coded with phase w.sub.k=2π(k−1)/M.sub.t, which are mapped to M.sub.t Doppler subbands (where k is the index of M.sub.t). Diagram **1650** illustrates an example of slow time phase coding for three SRS antenna ports.

[0163] In some designs, there may be different levels of DDM for SRS multiplexing. In this case, there may be sub-symbol level DDM, symbol level DDM, or resource level DDM. In sub-symbol level DDM, each SRS symbol may include multiple FMCWs. For example, comb-4 SRS will result in four FMCW waveforms, where these four FMCW waveforms may be further multiplexed through DDM. As such, with the comb structure, there may be repeated multiple FMCWs in the time domain. These repeated FMCWs arc transmitted within the same symbol and DDM is applied for these repeated FMCWs within the same symbol. For symbol level DDM, DDM is applied to SRS with the same comb offset. For resource level DDM, all of the SRS symbols for each antenna port are multiplied with a common phase code.

[0164] FIGS. **17**A and **17**B illustrate an example of resource level DDM, according to aspects of the disclosure. Diagram **1700** illustrates an example of slow time phase coding per resource (here, SRS resources **1** to N). Diagram **1750** illustrates an example of slow time phase coding per resource group. In this case, there may be several consecutive SRS resources bundled together and multiplied with a common phase code.

[0165] FIG. **18** illustrates an example method **1800** of wireless communication, according to aspects of the disclosure. In an aspect, method **1800** may be performed by a UE (e.g., any of the UEs described herein).

[0166] At **1810**, the UE transmits one or more SRS resources, wherein a waveform of the one or more SRS resources is an OFDM compatible FMCW waveform. In an aspect, operation **1810** may be performed by the one or more WWAN transceivers **310**, the one or more short-range wireless transceivers **320**, the one or more processors **342**, memory **340**, and/or positioning component **348**, any or all of which may be considered means for performing this operation.

[0167] As will be appreciated, a technical advantage of the method **1800** is improved uplink communication coverage, positioning and sensing capacity, and power consumption.

[0168] FIG. **19** illustrates an example method **1900** of communication, according to aspects of the disclosure. In an aspect, method **1900** may be performed by a network entity (e.g., an LMF **270**, a base station, or the like).

[0169] At **1910**, the network entity transmits, to a UE (e.g., any of the UEs described herein), a request for capabilities of the UE related to transmission of SRS resources by the UE. In an aspect, where the network entity is a server or other core network entity, operation **1910** may be performed by the one or more network transceivers **390**, the one or more processors **394**, memory **396**, and/or positioning component **398**, any or all of which may be considered means for performing this operation. In an aspect, where the network entity is a base station, operation **1910** may be performed by the one or more WWAN transceivers **350**, the one or more short-range wireless transceivers **360**, the one or more processors **384**, memory **386**, and/or positioning component **388**, any or all of which may be considered means for performing this operation.

[0170] At **1920**, the network entity receives, from the UE, one or more capabilities of the UE in response to the request, wherein the one or more capabilities of the UE indicate whether the UE supports an OFDM compatible FMCW waveform for the transmission of SRS resources. In an aspect, where the network entity is a server or other core network entity, operation **1920** may be performed by the one or more network transceivers **390**, the one or more processors **394**, memory **396**, and/or positioning component **398**, any or all of which may be considered means for performing this operation. In an aspect, where the network entity is a base station, operation **1920** may be performed by the one or more WWAN transceivers **350**, the one or more short-range wireless transceivers **360**, the one or more processors **384**, memory **386**, and/or positioning component **388**, any or all of which may be considered means for performing this operation.

[0171] As will be appreciated, a technical advantage of the method **1900** is enabling the network to determine the capabilities of the UE to support an OFDM compatible FMCW waveform for the transmission of SRS resources.

[0172] In the detailed description above it can be seen that different features are grouped together in examples. This manner of disclosure should not be understood as an intention that the example clauses have more features than are explicitly mentioned in each clause. Rather, the various aspects of the disclosure may include fewer than all features of an individual example clause disclosed. Therefore, the following clauses should hereby be deemed to be incorporated in the description, wherein each clause by itself can stand as a separate example. Although each dependent clause can refer in the clauses to a specific combination with one of the other clauses, the aspect(s) of that dependent clause are not limited to the specific combination. It will be appreciated that other example clauses can also include a combination of the dependent clause aspect(s) with the subject matter of any other dependent clause or independent clause or a combination of any feature with other dependent and independent clauses. The various aspects disclosed herein expressly include these combinations, unless it is explicitly expressed or can be readily inferred that a specific combination is not intended (e.g., contradictory aspects, such as defining an element as both an electrical insulator and an electrical conductor). Furthermore, it is also intended that aspects of a clause can be included in any other independent clause, even if the clause is not directly dependent on the independent clause.

[0173] Implementation examples are described in the following numbered clauses:

[0174] Clause 1. A method of wireless communication performed by a user equipment (UE), comprising: transmitting one or more sounding reference signal (SRS) resources, wherein a waveform of the one or more SRS resources is an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform.

[0175] Clause 2. The method of clause 1, wherein a configuration of the one or more SRS resources is based on one or more capabilities of the UE.

[0176] Clause 3. The method of clause 2, wherein the one or more capabilities of the UE indicate: whether the UE supports OFDM compatible FMCW waveforms, whether the UE supports multiplexing between FMCW transmissions and OFDM transmissions, whether the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, whether the UE supports gap symbols between SRS symbols, or any combination thereof.

[0177] Clause 4. The method of clause 3, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW waveforms, and the one or more capabilities further indicate one or more types of OFDM compatible FMCW waveforms the UE supports.

[0178] Clause 5. The method of clause 4, wherein the one or more types of OFDM compatible FMCW waveforms the UE supports include: zero-tail FMCW waveforms, piecewise linear chirp with cyclic prefix structure waveforms, or any combination thereof.

[0179] Clause 6. The method of any of clauses 3 to 5, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, and the one or more capabilities further indicate whether the UE supports different comb offsets in adjacent OFDM symbols for SRS transmission.

[0180] Clause 7. The method of any of clauses 3 to 6, wherein: the one or more capabilities indicate that the UE does not support OFDM compatible FMCW transmission with different comb offsets in adjacent symbols, and symbols of the one or more SRS resources are separated by one or more gap symbols.

[0181] Clause 8. The method of clause 7, wherein the one or more gap symbols are used for transmission by other UEs, other uplink channels, or both.

[0182] Clause 9. The method of any of clauses 2 to 8, wherein the one or more capabilities of the UE indicate: whether the UE supports a larger bandwidth part (BWP) for SRS transmission than a baseband BWP of the UE, whether the UE supports concurrent uplink transmissions using the larger BWP and the baseband BWP, or any combination thereof.

[0183] Clause 10. The method of clause 9, wherein: the one or more capabilities of the UE indicate that the UE does not support concurrent uplink transmissions using the larger BWP and the baseband BWP, and the one or more capabilities of the UE indicate a switching time the UE needs between SRS transmissions using the larger BWP and uplink transmissions using the baseband BWP.

[0184] Clause 11. The method of any of clauses 2 to 10, further comprising: receiving, from a network entity, a request for the one or more capabilities; and transmitting, to the network entity, the one or more capabilities of the UE in response to the request.

[0185] Clause 12. The method of any of clauses 1 to 11, wherein the one or more SRS resources are transmitted: over a larger BWP than a baseband BWP of the UE, or during one or more measurement gaps for SRS transmission.

[0186] Clause 13. The method of clause 12, wherein only SRS are transmitted over the larger BWP

and other uplink channels are transmitted over the baseband BWP.
[0187] Clause 14. The method of any of clauses 1 to 13, wherein: the one or more SRS resources comprise at least two SRS resources, the at least two SRS resources occupy the same OFDM resources, and the at least two SRS resources are Doppler-division multiplexed on the same OFDM resources.
[0188] Clause 15. The method of any of clauses 1 to 14, wherein: each of the one or more SRS resources comprises a plurality of FMCWs corresponding to a comb size of the one or more SRS resources, and the plurality of FMCWs is multiplexed using Doppler-division multiplexing (DDM).
[0189] Clause 16. The method of clause 15, wherein: the plurality of FMCWs is multiplexed at a symbol level, where symbols of the one or more SRS resources having the same comb offset are multiplexed using DDM, the plurality of FMCWs is multiplexed at a resource level, where all SRS symbols for each antenna port are multiplied with a common phase code, or the plurality of FMCWs is multiplexed at a sub-symbol level, where each SRS symbol of the one or more SRS resources includes multiple FMCWs of the plurality of FMCWs.
[0190] Clause 17. The method of any of clauses 1 to 16, wherein the OFDM compatible FMCW waveform is: a zero-tail FMCW waveform, or a piecewise linear chirp with cyclic prefix structure waveform.
[0191] Clause 18. The method of any of clauses 1 to 17, wherein: the one or more SRS resources comprise a plurality of resource elements, and the OFDM compatible FMCW is transmitted in each resource element of the plurality of resource elements.
[0192] Clause 19. A method of communication performed by a network entity, comprising: transmitting, to a user equipment (UE), a request for capabilities of the UE related to transmission of sounding reference signal (SRS) resources by the UE; and receiving, from the UE, one or more capabilities of the UE in response to the request, wherein the one or more capabilities of the UE indicate whether the UE supports an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform for the transmission of SRS resources.
[0193] Clause 20. The method of clause 19, wherein the one or more capabilities of the UE indicate: whether the UE supports OFDM compatible FMCW waveforms, whether the UE supports multiplexing between FMCW transmissions and OFDM transmissions, whether the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, whether the UE supports gap symbols between SRS symbols, or any combination thereof.
[0194] Clause 21. The method of clause 20, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW waveforms, and the one or more capabilities further indicate one or more types of OFDM compatible FMCW waveforms the UE supports.
[0195] Clause 22. The method of any of clauses 20 to 21, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, and the one or more capabilities further indicate whether the UE supports different comb offsets in adjacent OFDM symbols for SRS transmission.
[0196] Clause 23. The method of any of clauses 20 to 22, wherein: the one or more capabilities indicate that the UE does not support OFDM compatible FMCW transmission with different comb offsets in adjacent symbols, and symbols of SRS resources transmitted by the UE are separated by one or more gap symbols.
[0197] Clause 24. The method of any of clauses 19 to 23, wherein the one or more capabilities of the UE indicate: whether the UE supports a larger bandwidth part (BWP) for SRS transmission than a baseband BWP of the UE, whether the UE supports concurrent uplink transmissions using the larger BWP and the baseband BWP, or any combination thereof.
[0198] Clause 25. The method of clause 24, wherein: the one or more capabilities of the UE indicate that the UE does not support concurrent uplink transmissions using the larger BWP and the baseband BWP, and the one or more capabilities of the UE indicate a switching time the UE needs

between SRS transmissions using the larger BWP and uplink transmissions using the baseband BWP.
[0199] Clause 26. The method of any of clauses 19 to 25, wherein the transmission of SRS resources is: over a larger BWP than a baseband BWP of the UE, or during one or more measurement gaps for SRS transmission.
[0200] Clause 27. The method of any of clauses 19 to 26, wherein the OFDM compatible FMCW waveform is: a zero-tail FMCW waveform, or a piecewise linear chirp with cyclic prefix structure waveform.
[0201] Clause 28. The method of any of clauses 19 to 27, wherein: SRS resources transmitted by the UE comprise a plurality of resource elements, and the OFDM compatible FMCW is transmitted in each resource element of the plurality of resource elements.
[0202] Clause 29. A user equipment (UE), comprising: one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: transmit, via the one or more transceivers, one or more sounding reference signal (SRS) resources, wherein a waveform of the one or more SRS resources is an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform.
[0203] Clause 30. The UE of clause 29, wherein a configuration of the one or more SRS resources is based on one or more capabilities of the UE.
[0204] Clause 31. The UE of clause 30, wherein the one or more capabilities of the UE indicate: whether the UE supports OFDM compatible FMCW waveforms, whether the UE supports multiplexing between FMCW transmissions and OFDM transmissions, whether the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, whether the UE supports gap symbols between SRS symbols, or any combination thereof.
[0205] Clause 32. The UE of clause 31, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW waveforms, and the one or more capabilities further indicate one or more types of OFDM compatible FMCW waveforms the UE supports.
[0206] Clause 33. The UE of clause 32, wherein the one or more types of OFDM compatible FMCW waveforms the UE supports include: zero-tail FMCW waveforms, piecewise linear chirp with cyclic prefix structure waveforms, or any combination thereof.
[0207] Clause 34. The UE of any of clauses 31 to 33, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, and the one or more capabilities further indicate whether the UE supports different comb offsets in adjacent OFDM symbols for SRS transmission.
[0208] Clause 35. The UE of any of clauses 31 to 34, wherein: the one or more capabilities indicate that the UE does not support OFDM compatible FMCW transmission with different comb offsets in adjacent symbols, and symbols of the one or more SRS resources are separated by one or more gap symbols.
[0209] Clause 36. The UE of clause 35, wherein the one or more gap symbols are used for transmission by other UEs, other uplink channels, or both.
[0210] Clause 37. The UE of any of clauses 30 to 36, wherein the one or more capabilities of the UE indicate: whether the UE supports a larger bandwidth part (BWP) for SRS transmission than a baseband BWP of the UE, whether the UE supports concurrent uplink transmissions using the larger BWP and the baseband BWP, or any combination thereof.
[0211] Clause 38. The UE of clause 37, wherein: the one or more capabilities of the UE indicate that the UE does not support concurrent uplink transmissions using the larger BWP and the baseband BWP, and the one or more capabilities of the UE indicate a switching time the UE needs between SRS transmissions using the larger BWP and uplink transmissions using the baseband BWP.

[0212] Clause 39. The UE of any of clauses 30 to 38, wherein the one or more processors, either alone or in combination, are further configured to: receive, via the one or more transceivers, from a network entity, a request for the one or more capabilities; and transmit, via the one or more transceivers, to the network entity, the one or more capabilities of the UE in response to the request.
[0213] Clause 40. The UE of any of clauses 29 to 39, wherein the one or more SRS resources are transmitted: over a larger BWP than a baseband BWP of the UE, or during one or more measurement gaps for SRS transmission.
[0214] Clause 41. The UE of clause 40, wherein only SRS are transmitted over the larger BWP and other uplink channels are transmitted over the baseband BWP.
[0215] Clause 42. The UE of any of clauses 29 to 41, wherein: the one or more SRS resources comprise at least two SRS resources, the at least two SRS resources occupy the same OFDM resources, and the at least two SRS resources are Doppler-division multiplexed on the same OFDM resources.
[0216] Clause 43. The UE of any of clauses 29 to 42, wherein: each of the one or more SRS resources comprises a plurality of FMCWs corresponding to a comb size of the one or more SRS resources, and the plurality of FMCWs is multiplexed using Doppler-division multiplexing (DDM).
[0217] Clause 44. The UE of clause 43, wherein: the plurality of FMCWs is multiplexed at a symbol level, where symbols of the one or more SRS resources having the same comb offset are multiplexed using DDM, the plurality of FMCWs is multiplexed at a resource level, where all SRS symbols for each antenna port are multiplied with a common phase code, or the plurality of FMCWs is multiplexed at a sub-symbol level, where each SRS symbol of the one or more SRS resources includes multiple FMCWs of the plurality of FMCWs.
[0218] Clause 45. The UE of any of clauses 29 to 44, wherein the OFDM compatible FMCW waveform is: a zero-tail FMCW waveform, or a piecewise linear chirp with cyclic prefix structure waveform.
[0219] Clause 46. The UE of any of clauses 29 to 45, wherein: the one or more SRS resources comprise a plurality of resource elements, and the OFDM compatible FMCW is transmitted in each resource element of the plurality of resource elements.
[0220] Clause 47. A network entity, comprising: one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: transmit, via the one or more transceivers, to a user equipment (UE), a request for capabilities of the UE related to transmission of sounding reference signal (SRS) resources by the UE; and receive, via the one or more transceivers, from the UE, one or more capabilities of the UE in response to the request, wherein the one or more capabilities of the UE indicate whether the UE supports an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform for the transmission of SRS resources.
[0221] Clause 48. The network entity of clause 47, wherein the one or more capabilities of the UE indicate: whether the UE supports OFDM compatible FMCW waveforms, whether the UE supports multiplexing between FMCW transmissions and OFDM transmissions, whether the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, whether the UE supports gap symbols between SRS symbols, or any combination thereof.
[0222] Clause 49. The network entity of clause 48, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW waveforms, and the one or more capabilities further indicate one or more types of OFDM compatible FMCW waveforms the UE supports.
[0223] Clause 50. The network entity of any of clauses 48 to 49, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, and the one or more capabilities further indicate whether the UE supports different comb offsets in adjacent OFDM symbols for SRS transmission.
[0224] Clause 51. The network entity of any of clauses 48 to 50, wherein: the one or more

capabilities indicate that the UE does not support OFDM compatible FMCW transmission with different comb offsets in adjacent symbols, and symbols of SRS resources transmitted by the UE are separated by one or more gap symbols.

[0225] Clause 52. The network entity of any of clauses 47 to 51, wherein the one or more capabilities of the UE indicate: whether the UE supports a larger bandwidth part (BWP) for SRS transmission than a baseband BWP of the UE, whether the UE supports concurrent uplink transmissions using the larger BWP and the baseband BWP, or any combination thereof.

[0226] Clause 53. The network entity of clause 52, wherein: the one or more capabilities of the UE indicate that the UE does not support concurrent uplink transmissions using the larger BWP and the baseband BWP, and the one or more capabilities of the UE indicate a switching time the UE needs between SRS transmissions using the larger BWP and uplink transmissions using the baseband BWP.

[0227] Clause 54. The network entity of any of clauses 47 to 53, wherein the transmission of SRS resources is: over a larger BWP than a baseband BWP of the UE, or during one or more measurement gaps for SRS transmission.

[0228] Clause 55. The network entity of any of clauses 47 to 54, wherein the OFDM compatible FMCW waveform is: a zero-tail FMCW waveform, or a piecewise linear chirp with cyclic prefix structure waveform.

[0229] Clause 56. The network entity of any of clauses 47 to 55, wherein: SRS resources transmitted by the UE comprise a plurality of resource elements, and the OFDM compatible FMCW is transmitted in each resource element of the plurality of resource elements.

[0230] Clause 57. A user equipment (UE), comprising: means for transmitting one or more sounding reference signal (SRS) resources, wherein a waveform of the one or more SRS resources is an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform.

[0231] Clause 58. The UE of clause 57, wherein a configuration of the one or more SRS resources is based on one or more capabilities of the UE.

[0232] Clause 59. The UE of clause 58, wherein the one or more capabilities of the UE indicate: whether the UE supports OFDM compatible FMCW waveforms, whether the UE supports multiplexing between FMCW transmissions and OFDM transmissions, whether the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, whether the UE supports gap symbols between SRS symbols, or any combination thereof.

[0233] Clause 60. The UE of clause 59, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW waveforms, and the one or more capabilities further indicate one or more types of OFDM compatible FMCW waveforms the UE supports.

[0234] Clause 61. The UE of clause 60, wherein the one or more types of OFDM compatible FMCW waveforms the UE supports include: zero-tail FMCW waveforms, piecewise linear chirp with cyclic prefix structure waveforms, or any combination thereof.

[0235] Clause 62. The UE of any of clauses 59 to 61, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, and the one or more capabilities further indicate whether the UE supports different comb offsets in adjacent OFDM symbols for SRS transmission.

[0236] Clause 63. The UE of any of clauses 59 to 62, wherein: the one or more capabilities indicate that the UE does not support OFDM compatible FMCW transmission with different comb offsets in adjacent symbols, and symbols of the one or more SRS resources are separated by one or more gap symbols.

[0237] Clause 64. The UE of clause 63, wherein the one or more gap symbols are used for transmission by other UEs, other uplink channels, or both.

[0238] Clause 65. The UE of any of clauses 58 to 64, wherein the one or more capabilities of the UE indicate: whether the UE supports a larger bandwidth part (BWP) for SRS transmission than a baseband BWP of the UE, whether the UE supports concurrent uplink transmissions using the larger BWP and the baseband BWP, or any combination thereof.

[0239] Clause 66. The UE of clause 65, wherein: the one or more capabilities of the UE indicate that the UE does not support concurrent uplink transmissions using the larger BWP and the baseband BWP, and the one or more capabilities of the UE indicate a switching time the UE needs between SRS transmissions using the larger BWP and uplink transmissions using the baseband BWP.

[0240] Clause 67. The UE of any of clauses 58 to 66, further comprising: means for receiving, from a network entity, a request for the one or more capabilities; and means for transmitting, to the network entity, the one or more capabilities of the UE in response to the request.

[0241] Clause 68. The UE of any of clauses 57 to 67, wherein the one or more SRS resources are transmitted: over a larger BWP than a baseband BWP of the UE, or during one or more measurement gaps for SRS transmission.

[0242] Clause 69. The UE of clause 68, wherein only SRS are transmitted over the larger BWP and other uplink channels are transmitted over the baseband BWP.

[0243] Clause 70. The UE of any of clauses 57 to 69, wherein: the one or more SRS resources comprise at least two SRS resources, the at least two SRS resources occupy the same OFDM resources, and the at least two SRS resources are Doppler-division multiplexed on the same OFDM resources.

[0244] Clause 71. The UE of any of clauses 57 to 70, wherein: each of the one or more SRS resources comprises a plurality of FMCWs corresponding to a comb size of the one or more SRS resources, and the plurality of FMCWs is multiplexed using Doppler-division multiplexing (DDM).

[0245] Clause 72. The UE of clause 71, wherein: the plurality of FMCWs is multiplexed at a symbol level, where symbols of the one or more SRS resources having the same comb offset are multiplexed using DDM, the plurality of FMCWs is multiplexed at a resource level, where all SRS symbols for each antenna port are multiplied with a common phase code, or the plurality of FMCWs is multiplexed at a sub-symbol level, where each SRS symbol of the one or more SRS resources includes multiple FMCWs of the plurality of FMCWs.

[0246] Clause 73. The UE of any of clauses 57 to 72, wherein the OFDM compatible FMCW waveform is: a zero-tail FMCW waveform, or a piecewise linear chirp with cyclic prefix structure waveform.

[0247] Clause 74. The UE of any of clauses 57 to 73, wherein: the one or more SRS resources comprise a plurality of resource elements, and the OFDM compatible FMCW is transmitted in each resource element of the plurality of resource elements.

[0248] Clause 75. A network entity, comprising: means for transmitting, to a user equipment (UE), a request for capabilities of the UE related to transmission of sounding reference signal (SRS) resources by the UE; and means for receiving, from the UE, one or more capabilities of the UE in response to the request, wherein the one or more capabilities of the UE indicate whether the UE supports an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform for the transmission of SRS resources.

[0249] Clause 76. The network entity of clause 75, wherein the one or more capabilities of the UE indicate: whether the UE supports OFDM compatible FMCW waveforms, whether the UE supports multiplexing between FMCW transmissions and OFDM transmissions, whether the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, whether the UE supports gap symbols between SRS symbols, or any combination thereof.

[0250] Clause 77. The network entity of clause 76, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW waveforms, and the one or more capabilities further indicate one or more types of OFDM compatible FMCW waveforms the UE supports.

[0251] Clause 78. The network entity of any of clauses 76 to 77, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW transmission with different

comb offsets in different symbols, and the one or more capabilities further indicate whether the UE supports different comb offsets in adjacent OFDM symbols for SRS transmission.
[0252] Clause 79. The network entity of any of clauses 76 to 78, wherein: the one or more capabilities indicate that the UE does not support OFDM compatible FMCW transmission with different comb offsets in adjacent symbols, and symbols of SRS resources transmitted by the UE are separated by one or more gap symbols.
[0253] Clause 80. The network entity of any of clauses 75 to 79, wherein the one or more capabilities of the UE indicate: whether the UE supports a larger bandwidth part (BWP) for SRS transmission than a baseband BWP of the UE, whether the UE supports concurrent uplink transmissions using the larger BWP and the baseband BWP, or any combination thereof.
[0254] Clause 81. The network entity of clause 80, wherein: the one or more capabilities of the UE indicate that the UE does not support concurrent uplink transmissions using the larger BWP and the baseband BWP, and the one or more capabilities of the UE indicate a switching time the UE needs between SRS transmissions using the larger BWP and uplink transmissions using the baseband BWP.
[0255] Clause 82. The network entity of any of clauses 75 to 81, wherein the transmission of SRS resources is: over a larger BWP than a baseband BWP of the UE, or during one or more measurement gaps for SRS transmission.
[0256] Clause 83. The network entity of any of clauses 75 to 82, wherein the OFDM compatible FMCW waveform is: a zero-tail FMCW waveform, or a piecewise linear chirp with cyclic prefix structure waveform.
[0257] Clause 84. The network entity of any of clauses 75 to 83, wherein: SRS resources transmitted by the UE comprise a plurality of resource elements, and the OFDM compatible FMCW is transmitted in each resource element of the plurality of resource elements.
[0258] Clause 85. A non-transitory computer-readable medium storing computer-executable instructions that, when executed by a user equipment (UE), cause the UE to: transmit one or more sounding reference signal (SRS) resources, wherein a waveform of the one or more SRS resources is an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform.
[0259] Clause 86. The non-transitory computer-readable medium of clause 85, wherein a configuration of the one or more SRS resources is based on one or more capabilities of the UE.
[0260] Clause 87. The non-transitory computer-readable medium of clause 86, wherein the one or more capabilities of the UE indicate: whether the UE supports OFDM compatible FMCW waveforms, whether the UE supports multiplexing between FMCW transmissions and OFDM transmissions, whether the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, whether the UE supports gap symbols between SRS symbols, or any combination thereof.
[0261] Clause 88. The non-transitory computer-readable medium of clause 87, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW waveforms, and the one or more capabilities further indicate one or more types of OFDM compatible FMCW waveforms the UE supports.
[0262] Clause 89. The non-transitory computer-readable medium of clause 88, wherein the one or more types of OFDM compatible FMCW waveforms the UE supports include: zero-tail FMCW waveforms, piecewise linear chirp with cyclic prefix structure waveforms, or any combination thereof.
[0263] Clause 90. The non-transitory computer-readable medium of any of clauses 87 to 89, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, and the one or more capabilities further indicate whether the UE supports different comb offsets in adjacent OFDM symbols for SRS transmission.

[0264] Clause 91. The non-transitory computer-readable medium of any of clauses 87 to 90, wherein: the one or more capabilities indicate that the UE does not support OFDM compatible FMCW transmission with different comb offsets in adjacent symbols, and symbols of the one or more SRS resources are separated by one or more gap symbols.
[0265] Clause 92. The non-transitory computer-readable medium of clause 91, wherein the one or more gap symbols are used for transmission by other UEs, other uplink channels, or both.
[0266] Clause 93. The non-transitory computer-readable medium of any of clauses 86 to 92, wherein the one or more capabilities of the UE indicate: whether the UE supports a larger bandwidth part (BWP) for SRS transmission than a baseband BWP of the UE, whether the UE supports concurrent uplink transmissions using the larger BWP and the baseband BWP, or any combination thereof.
[0267] Clause 94. The non-transitory computer-readable medium of clause 93, wherein: the one or more capabilities of the UE indicate that the UE does not support concurrent uplink transmissions using the larger BWP and the baseband BWP, and the one or more capabilities of the UE indicate a switching time the UE needs between SRS transmissions using the larger BWP and uplink transmissions using the baseband BWP.
[0268] Clause 95. The non-transitory computer-readable medium of any of clauses 86 to 94, further comprising computer-executable instructions that, when executed by the UE, cause the UE to: receive, from a network entity, a request for the one or more capabilities; and transmit, to the network entity, the one or more capabilities of the UE in response to the request.
[0269] Clause 96. The non-transitory computer-readable medium of any of clauses 85 to 95, wherein the one or more SRS resources are transmitted: over a larger BWP than a baseband BWP of the UE, or during one or more measurement gaps for SRS transmission.
[0270] Clause 97. The non-transitory computer-readable medium of clause 96, wherein only SRS are transmitted over the larger BWP and other uplink channels are transmitted over the baseband BWP.
[0271] Clause 98. The non-transitory computer-readable medium of any of clauses 85 to 97, wherein: the one or more SRS resources comprise at least two SRS resources, the at least two SRS resources occupy the same OFDM resources, and the at least two SRS resources are Doppler-division multiplexed on the same OFDM resources.
[0272] Clause 99. The non-transitory computer-readable medium of any of clauses 85 to 98, wherein: each of the one or more SRS resources comprises a plurality of FMCWs corresponding to a comb size of the one or more SRS resources, and the plurality of FMCWs is multiplexed using Doppler-division multiplexing (DDM).
[0273] Clause 100. The non-transitory computer-readable medium of clause 99, wherein: the plurality of FMCWs is multiplexed at a symbol level, where symbols of the one or more SRS resources having the same comb offset are multiplexed using DDM, the plurality of FMCWs is multiplexed at a resource level, where all SRS symbols for each antenna port are multiplied with a common phase code, or the plurality of FMCWs is multiplexed at a sub-symbol level, where each SRS symbol of the one or more SRS resources includes multiple FMCWs of the plurality of FMCWs.
[0274] Clause 101. The non-transitory computer-readable medium of any of clauses 85 to 100, wherein the OFDM compatible FMCW waveform is: a zero-tail FMCW waveform, or a piecewise linear chirp with cyclic prefix structure waveform.
[0275] Clause 102. The non-transitory computer-readable medium of any of clauses 85 to 101, wherein: the one or more SRS resources comprise a plurality of resource elements, and the OFDM compatible FMCW is transmitted in each resource element of the plurality of resource elements.
[0276] Clause 103. A non-transitory computer-readable medium storing computer-executable instructions that, when executed by a network entity, cause the network entity to: transmit, to a user equipment (UE), a request for capabilities of the UE related to transmission of sounding reference

signal (SRS) resources by the UE; and receive, from the UE, one or more capabilities of the UE in response to the request, wherein the one or more capabilities of the UE indicate whether the UE supports an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform for the transmission of SRS resources.

[0277] Clause 104. The non-transitory computer-readable medium of clause 103, wherein the one or more capabilities of the UE indicate: whether the UE supports OFDM compatible FMCW waveforms, whether the UE supports multiplexing between FMCW transmissions and OFDM transmissions, whether the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, whether the UE supports gap symbols between SRS symbols, or any combination thereof.

[0278] Clause 105. The non-transitory computer-readable medium of clause 104, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW waveforms, and the one or more capabilities further indicate one or more types of OFDM compatible FMCW waveforms the UE supports.

[0279] Clause 106. The non-transitory computer-readable medium of any of clauses 104 to 105, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, and the one or more capabilities further indicate whether the UE supports different comb offsets in adjacent OFDM symbols for SRS transmission.

[0280] Clause 107. The non-transitory computer-readable medium of any of clauses 104 to 106, wherein: the one or more capabilities indicate that the UE does not support OFDM compatible FMCW transmission with different comb offsets in adjacent symbols, and symbols of SRS resources transmitted by the UE are separated by one or more gap symbols.

[0281] Clause 108. The non-transitory computer-readable medium of any of clauses 103 to 107, wherein the one or more capabilities of the UE indicate: whether the UE supports a larger bandwidth part (BWP) for SRS transmission than a baseband BWP of the UE, whether the UE supports concurrent uplink transmissions using the larger BWP and the baseband BWP, or any combination thereof.

[0282] Clause 109. The non-transitory computer-readable medium of clause 108, wherein: the one or more capabilities of the UE indicate that the UE does not support concurrent uplink transmissions using the larger BWP and the baseband BWP, and the one or more capabilities of the UE indicate a switching time the UE needs between SRS transmissions using the larger BWP and uplink transmissions using the baseband BWP.

[0283] Clause 110. The non-transitory computer-readable medium of any of clauses 103 to 109, wherein the transmission of SRS resources is: over a larger BWP than a baseband BWP of the UE, or during one or more measurement gaps for SRS transmission.

[0284] Clause 111. The non-transitory computer-readable medium of any of clauses 103 to 110, wherein the OFDM compatible FMCW waveform is: a zero-tail FMCW waveform, or a piecewise linear chirp with cyclic prefix structure waveform.

[0285] Clause 112. The non-transitory computer-readable medium of any of clauses 103 to 111, wherein: SRS resources transmitted by the UE comprise a plurality of resource elements, and the OFDM compatible FMCW is transmitted in each resource element of the plurality of resource elements.

[0286] Those of skill in the art will appreciate that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0287] Further, those of skill in the art will appreciate that the various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

[0288] The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an ASIC, a field-programable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, for example, a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0289] The methods, sequences and/or algorithms described in connection with the aspects disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in random access memory (RAM), flash memory, read-only memory (ROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), registers, hard disk, a removable disk, a CD-ROM, or any other form of storage medium known in the art. An example storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a user terminal (e.g., UE). In the alternative, the processor and the storage medium may reside as discrete components in a user terminal.

[0290] In one or more example aspects, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A storage media may be any available media that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code in the form of instructions or data structures and that can be accessed by a computer. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

[0291] While the foregoing disclosure shows illustrative aspects of the disclosure, it should be noted that various changes and modifications could be made herein without departing from the scope of the disclosure as defined by the appended claims. For example, the functions, steps and/or actions of the method claims in accordance with the aspects of the disclosure described herein need not be performed in any particular order. Further, no component, function, action, or instruction described or claimed herein should be construed as critical or essential unless explicitly described as such. Furthermore, as used herein, the terms “set,” “group,” and the like are intended to include one or more of the stated elements. Also, as used herein, the terms “has,” “have,” “having,” “comprises,” “comprising,” “includes,” “including,” and the like does not preclude the presence of one or more additional elements (e.g., an element “having” A may also have B). Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”) or the alternatives are mutually exclusive (e.g., “one or more” should not be interpreted as “one and more”). Furthermore, although components, functions, actions, and instructions may be described or claimed in the singular, the plural is contemplated unless limitation to the singular is explicitly stated. Accordingly, as used herein, the articles “a,” “an,” “the,” and “said” are intended to include one or more of the stated elements. Additionally, as used herein, the terms “at least one” and “one or more” encompass “one” component, function, action, or instruction performing or capable of performing a described or claimed functionality and also “two or more” components, functions, actions, or instructions performing or capable of performing a described or claimed functionality in combination.

Claims

1. A user equipment (UE), comprising: one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: transmit, via the one or more transceivers, one or more sounding reference signal (SRS) resources, wherein a waveform of the one or more SRS resources is an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform.
2. The UE of claim 1, wherein a configuration of the one or more SRS resources is based on one or more capabilities of the UE.
3. The UE of claim 2, wherein the one or more capabilities of the UE indicate: whether the UE supports OFDM compatible FMCW waveforms, whether the UE supports multiplexing between FMCW transmissions and OFDM transmissions, whether the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, whether the UE supports gap symbols between SRS symbols, or any combination thereof.
4. The UE of claim 3, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW waveforms, and the one or more capabilities further indicate one or more types of OFDM compatible FMCW waveforms the UE supports.
5. The UE of claim 4, wherein the one or more types of OFDM compatible FMCW waveforms the UE supports include: zero-tail FMCW waveforms, piecewise linear chirp with cyclic prefix structure waveforms, or any combination thereof.
6. The UE of claim 3, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, and the one or more capabilities further indicate whether the UE supports different comb offsets in adjacent OFDM symbols for SRS transmission.
7. The UE of claim 3, wherein: the one or more capabilities indicate that the UE does not support OFDM compatible FMCW transmission with different comb offsets in adjacent symbols, and symbols of the one or more SRS resources are separated by one or more gap symbols.
8. The UE of claim 7, wherein the one or more gap symbols are used for transmission by other

UEs, other uplink channels, or both.
9. The UE of claim 2, wherein the one or more capabilities of the UE indicate: whether the UE supports a larger bandwidth part (BWP) for SRS transmission than a baseband BWP of the UE, whether the UE supports concurrent uplink transmissions using the larger BWP and the baseband BWP, or any combination thereof.
10. The UE of claim 9, wherein: the one or more capabilities of the UE indicate that the UE does not support concurrent uplink transmissions using the larger BWP and the baseband BWP, and the one or more capabilities of the UE indicate a switching time the UE needs between SRS transmissions using the larger BWP and uplink transmissions using the baseband BWP.
11. The UE of claim 2, wherein the one or more processors, either alone or in combination, are further configured to: receive, via the one or more transceivers, from a network entity, a request for the one or more capabilities; and transmit, via the one or more transceivers, to the network entity, the one or more capabilities of the UE in response to the request.
12. The UE of claim 1, wherein the one or more SRS resources are transmitted: over a larger BWP than a baseband BWP of the UE, or during one or more measurement gaps for SRS transmission.
13. The UE of claim 12, wherein only SRS are transmitted over the larger BWP and other uplink channels are transmitted over the baseband BWP.
14. The UE of claim 1, wherein: the one or more SRS resources comprise at least two SRS resources, the at least two SRS resources occupy the same OFDM resources, and the at least two SRS resources are Doppler-division multiplexed on the same OFDM resources.
15. The UE of claim 1, wherein: each of the one or more SRS resources comprises a plurality of FMCWs corresponding to a comb size of the one or more SRS resources, and the plurality of FMCWs is multiplexed using Doppler-division multiplexing (DDM).
16. The UE of claim 15, wherein: the plurality of FMCWs is multiplexed at a symbol level, where symbols of the one or more SRS resources having the same comb offset are multiplexed using DDM, the plurality of FMCWs is multiplexed at a resource level, where all SRS symbols for each antenna port are multiplied with a common phase code, or the plurality of FMCWs is multiplexed at a sub-symbol level, where each SRS symbol of the one or more SRS resources includes multiple FMCWs of the plurality of FMCWs.
17. The UE of claim 1, wherein the OFDM compatible FMCW waveform is: a zero-tail FMCW waveform, or a piecewise linear chirp with cyclic prefix structure waveform.
18. The UE of claim 1, wherein: the one or more SRS resources comprise a plurality of resource elements, and the OFDM compatible FMCW is transmitted in each resource element of the plurality of resource elements.
19. A network entity, comprising: one or more memories; one or more transceivers; and one or more processors communicatively coupled to the one or more memories and the one or more transceivers, the one or more processors, either alone or in combination, configured to: transmit, via the one or more transceivers, to a user equipment (UE), a request for capabilities of the UE related to transmission of sounding reference signal (SRS) resources by the UE; and receive, via the one or more transceivers, from the UE, one or more capabilities of the UE in response to the request, wherein the one or more capabilities of the UE indicate whether the UE supports an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform for the transmission of SRS resources.
20. The network entity of claim 19, wherein the one or more capabilities of the UE indicate: whether the UE supports OFDM compatible FMCW waveforms, whether the UE supports multiplexing between FMCW transmissions and OFDM transmissions, whether the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, whether the UE supports gap symbols between SRS symbols, or any combination thereof.
21. The network entity of claim 20, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW waveforms, and the one or more capabilities further indicate

one or more types of OFDM compatible FMCW waveforms the UE supports.
22. The network entity of claim 20, wherein: the one or more capabilities indicate that the UE supports OFDM compatible FMCW transmission with different comb offsets in different symbols, and the one or more capabilities further indicate whether the UE supports different comb offsets in adjacent OFDM symbols for SRS transmission.
23. The network entity of claim 20, wherein: the one or more capabilities indicate that the UE does not support OFDM compatible FMCW transmission with different comb offsets in adjacent symbols, and symbols of SRS resources transmitted by the UE are separated by one or more gap symbols.
24. The network entity of claim 19, wherein the one or more capabilities of the UE indicate: whether the UE supports a larger bandwidth part (BWP) for SRS transmission than a baseband BWP of the UE, whether the UE supports concurrent uplink transmissions using the larger BWP and the baseband BWP, or any combination thereof.
25. The network entity of claim 24, wherein: the one or more capabilities of the UE indicate that the UE does not support concurrent uplink transmissions using the larger BWP and the baseband BWP, and the one or more capabilities of the UE indicate a switching time the UE needs between SRS transmissions using the larger BWP and uplink transmissions using the baseband BWP.
26. The network entity of claim 19, wherein the transmission of SRS resources is: over a larger BWP than a baseband BWP of the UE, or during one or more measurement gaps for SRS transmission.
27. The network entity of claim 19, wherein the OFDM compatible FMCW waveform is: a zero-tail FMCW waveform, or a piecewise linear chirp with cyclic prefix structure waveform.
28. The network entity of claim 19, wherein: SRS resources transmitted by the UE comprise a plurality of resource elements, and the OFDM compatible FMCW is transmitted in each resource element of the plurality of resource elements.
29. A method of wireless communication performed by a user equipment (UE), comprising: transmitting one or more sounding reference signal (SRS) resources, wherein a waveform of the one or more SRS resources is an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform.
30. A method of communication performed by a network entity, comprising: transmitting, to a user equipment (UE), a request for capabilities of the UE related to transmission of sounding reference signal (SRS) resources by the UE; and receiving, from the UE, one or more capabilities of the UE in response to the request, wherein the one or more capabilities of the UE indicate whether the UE supports an orthogonal frequency division multiplexing (OFDM) compatible frequency modulated continuous wave (FMCW) waveform for the transmission of SRS resources.